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Unbiased Reconstruction Network for Few-Shot Fine-Grained Remote Sensing Image Classification

Yangfan Li

***Abstract*—Reconstruction-based methods have achieved remarkable success in few-shot fine-grained remote sensing image classification by projecting support and query features into each other's spaces and measuring their similarity via reconstruction errors. Existing approaches typically follow a one-stage reconstruction paradigm, where each projection is conducted in a single step and optimized independently. However, this design leads to two critical limitations. First, the projection process may discard fine-grained discriminative details—particularly under low-rank or ill-conditioned mappings—resulting in representation bias. Second, although projection matrices are re-estimated on novel classes, the use of base-trained feature encoders induces a reconstruction bias, as the distributional shift inflates intra-class variance in novel-class embeddings and undermines the reliability of reconstruction-based metrics. To address these issues, we propose the Unbiased Reconstruction Network (URN), the first two-stage reconstruction framework that introduces a cyclic reconstruction objective with an inverse reconstruction constraint. Specifically, URN first projects support features into the query space and then reprojects the reconstructed query features back into the support space, enforcing bidirectional alignment and preserving structural semantics. This cyclic design mitigates both representation and reconstruction biases, leading to improved generalization on novel classes. Extensive experiments on multiple few-shot fine-grained remote sensing image classification benchmarks validate the effectiveness and superiority of URN over existing one-stage reconstruction methods. The code will be publicly released at https://github.com/liyangfan0/URN.**

***Index Terms*—Few-Shot Learning, Remote Sensing Image Classification, Feature Reconstruction.**

I. Introduction

FEW-shot fine-grained image classification is particularly challenging in remote sensing imagery, where distinguishing between visually similar ship types or complex scene categories requires subtle feature recognition with limited labeled data. This challenge is exacerbated in maritime monitoring, where fine-grained ship classification must account for varying environmental conditions, backgrounds, and scales. Similarly, scene classification in remote sensing images faces challenges due to the diversity in geographical features like forests, urban areas, and coastal regions. The scarcity of labeled data for rare ship types or specific scenes makes robust few-shot learning methods essential for effective classification in these contexts.

Few-shot fine-grained classification aims to learn transferable representations from a set of base classes with sufficient labeled samples and then recognize previously unseen novel classes using only a few labeled examples per class. The base and novel categories are non-overlapping, while the novel classes often exhibit subtle inter-class differences and large intra-class variations caused by changes in scale, viewpoint, background, and imaging conditions. Therefore, the task requires the model to achieve both fine-grained discrimination and rapid generalization under extremely limited supervision.

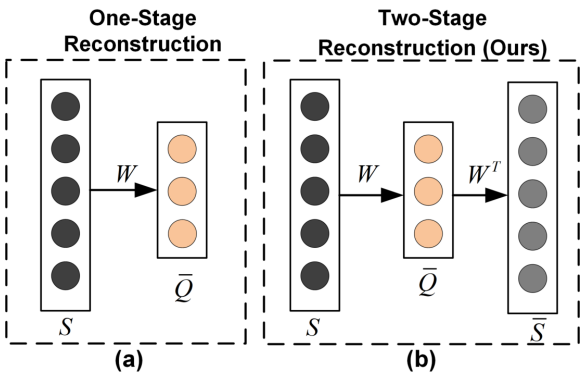


Fig. 1. Illustration of the reconstruction paradigms: (a) One-stage paradigm adopted by existing methods; (b) Two-stage paradigm proposed in URN. In (a), support features are directly projected into the query space, and similarity is computed via reconstruction error, which often leads to representation and reconstruction biases. In contrast, (b) shows the proposed two-stage reconstruction paradigm, where reconstructed query features are further mapped back to the support space with a cyclic constraint, enabling bidirectional alignment and reducing bias. This figure highlights the conceptual difference between the existing one-stage reconstruction paradigm and our two-stage reconstruction paradigm.

Under this setting, few-shot fine-grained remote sensing image classification faces two primary challenges. First, the differences between subclasses, such as ship types or scene categories, are often subtle, while intra-class variation is significant, requiring models with strong discriminative capabilities. Second, few-shot learning assumes that training and test categories are distinct, leading to a distribution shift between base and novel classes, making robust generalization essential, especially when dealing with complex remote sensing environments.

Recent advancements in few-shot image classification have primarily focused on three main approaches: global feature-based, local feature-based, and attention-based methods. Global feature-based methods [1], [2] aim to design distance metrics for global image features but often suffer from detail loss due to global pooling, limiting their effectiveness in fine-grained tasks like ship classification or scene recognition

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in remote sensing. On the other hand, local feature-based methods [3]–[14] focus on distance metrics between sets of feature descriptors, preserving rich detail information and achieving superior performance in fine-grained classification tasks. Attention-based methods [15]–[20] enhance performance by designing spatial or channel-wise weighting mechanisms to suppress irrelevant features, which can be particularly beneficial for isolating relevant features in complex remote sensing imagery.

Building on these paradigms, recent efforts in few-shot fine-grained classification have introduced more targeted designs to improve feature discriminability and mitigate intra-class ambiguity. Several state-of-the-art methods [6], [17], [20]–[27] propose strategies such as local feature enhancement, attention-based refinement, and discriminative loss optimization. For instance, FicNet [20] combines frequency-awareness with attention to refine discriminative feature learning, RaPSPNet [21] strengthens class separability via a feature re-abstraction module, and AIS-MLI [22] employs an Angular Isotonic Loss to enlarge inter-class margins, which can help enhance the classification of fine-grained ship types in remote sensing.

Among these methods, reconstruction-based approaches stand out as a conceptually elegant and empirically effective subfamily within the local feature-based paradigm. Representative techniques [6], [24]–[31] follow a general *projection–reconstruction–metric* pipeline: support and query features are projected into each other's spaces, reconstructed representations are generated, and reconstruction errors are used to measure similarity—smaller errors indicating greater correspondence. This design not only facilitates semantic alignment between support and query features but also offers flexibility for integrating architectural innovations and learning strategies, especially when dealing with complex, multi-scale features in remote sensing.

From a structural perspective, existing methods predominantly follow a one-stage reconstruction framework. FRN [6] performs a single support-to-query projection based on ridge regression and computes residuals for classification, forming a *one-path* architecture. BiFRN [24] and SRN [27] introduce an additional query-to-support projection, yielding a *two-path* setup. LCCRN [25] further decomposes features into global and local parts, leading to four separate cross-type projections, forming a *four-path* scheme. Despite the increasing number of projection paths, these methods still optimize each path independently, without coordination or mutual constraints, maintaining the one-stage paradigm.

However, this one-stage design gives rise to two fundamental limitations:

(1) *Representation bias.* Each projection is typically implemented as a single-step linear transformation, which tends to be low-rank or ill-conditioned in few-shot scenarios. This leads to the loss of fine-grained discriminative details during projection, which cannot be recovered—even by increasing the number of reconstruction paths or adding residual branches.

(2) *Reconstruction bias.* Although the projection matrix is re-estimated on novel classes, both support and query features are extracted by a base-trained encoder. Due to distributional shift, novel-class embeddings often exhibit inflated intra-class variance, violating the metric-learning assumption. This misalignment leads to unreliable reconstruction errors, degrading the model's ability to distinguish between same-class and different-class samples.

These insights reveal a critical bottleneck: simply increasing projection paths or combining more residuals cannot fundamentally overcome the dual challenges of *representation bias* and *reconstruction bias* in the one-stage paradigm. Existing multi-path methods mainly increase the *breadth* of reconstruction by introducing multiple independent projection paths, while each path still performs a single-stage reconstruction. In contrast, URN increases the *depth* of reconstruction by introducing a coupled forward–inverse reconstruction process, where the two mappings are jointly constrained through cyclic consistency rather than optimized independently.

To break this bottleneck, we propose the Unbiased Reconstruction Network (URN), the first *two-stage reconstruction* framework. Unlike prior multi-path methods, URN does not simply add additional reconstruction directions. Instead, it introduces an inverse reconstruction stage coupled with the forward reconstruction, forming a cyclic reconstruction mechanism between the support and query representations. This coupled two-stage design enforces mutual consistency between the forward and inverse mappings, thereby improving the reliability of feature reconstruction and similarity estimation (Figure 1).

To implement this idea, we design a novel *cyclic reconstruction objective* that explicitly constrains the forward and inverse projections. Solving this objective yields a more transferable and unbiased projection matrix. However, the resulting formulation leads to a Sylvester equation, which is computationally prohibitive to solve directly on large matrices using the Bartels-Stewart algorithm [32]. To address this challenge, we introduce an auxiliary virtual matrix and propose an alternating optimization strategy. The original objective is decomposed into two tractable sub-problems, each with a closed-form solution. This allows for efficient computation and fast empirical convergence within just a few iterations.

The main contributions of this paper are as follows:

- To address the issues of representation bias and reconstruction bias in current one-stage reconstruction methods, the first two-stage reconstruction framework—URN is proposed. By constructing a novel cyclic reconstruction objective function, URN implements an inverse reconstruction constraint, which not only ensures that the reconstructed query features retain rich fine-grained information from the support features but also improves the model's generalization ability to novel classes.
- To efficiently solve the cyclic reconstruction objective function in URN, an additional auxiliary matrix variable is introduced and the alternating optimization algorithm is presented, which decomposes the original optimization problem into two sub-problems that are solved iteratively, with each sub-problem having a closed-form solution. This approach results in extremely fast computation. Moreover, the algorithm can empirically converge within few iterations.

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- Extensive experiments are conducted on multiple few-shot fine-grained classification benchmarks. The results demonstrate that URN achieves state-of-the-art classification performance across all benchmarks, further proving the superiority of URN over existing one-stage reconstruction methods.

II. Related Work

A. Few-Shot Fine-Grained Image Classification

Few-shot fine-grained image classification (FSFGIC) aims to recognize novel categories from only a few labeled examples, while tackling small inter-class differences and large intra-class variations. Metric-based FSFGIC methods are generally grouped into local feature-based and attention-based methods:

Local feature-based methods preserve spatial feature maps and match dense descriptors. DeepEMD [12], [13] adopts Earth Mover's Distance; CrossTransformer [7] reconstructs queries using key–value subspaces; MCL [8] models bidirectional interactions. Other works like DeepBDC [10], FGFL [11], and CPEA [9] leverage statistical, frequency, or patch-specific cues. Recent works enhance fine-grained discrimination by modeling semantic relations [33], spatial structures [34], or structured metrics [35]. Other strategies include integrating frequency-aware features [20], re-abstracting support-query features to strengthen class separability [21], and introducing specialized loss functions to enlarge inter-class margins [22]. Among local feature-based methods, *reconstruction-based approaches* [6], [24]–[27] are notable for their projection–reconstruction–metric paradigm. By explicitly projecting support features into the query space and reconstructing the latter for comparison, these methods establish a structured interaction mechanism that goes beyond conventional similarity computation. Their distinctive formulation and theoretical underpinnings motivate a more in-depth examination, which we provide in the following section.

Attention-based methods enhance feature discrimination by adaptively focusing on informative spatial or channel regions, thus suppressing background noise and highlighting subtle inter-class differences. CAN [15] utilize cross-attention between support and query to localize discriminative regions. DAN [18] integrates both spatial and channel-wise attention to capture multi-dimensional cues. ReNet [16] models auto- and cross-correlation to better align relational patterns. TDM [17] reweights channels based on class-specific relevance. CAD [19] introduces cross-attention transformers for relation modeling based on transformer-based architectures. Further advancements explore task-adaptive feature learning [36] and geometric robustness, such as rotation-invariant modeling [37]. Despite their varying architectures, these methods share the goal of learning adaptive attention maps that generalize well across tasks with limited data.

B. Few-Shot Learning in Remote Sensing

In addition to general few-shot fine-grained methods in natural images, there has been increasing interest in few-shot learning for remote sensing image classification, particularly under limited labeled samples and cross-domain distribution shifts. Reconstruction-based techniques tailored for remote sensing, such as FAFRN [28], CWFRN [29], and RIFRN [30], have been proposed for fine-grained ship classification. Beyond object-level tasks, remote sensing scene classification has seen a variety of few-shot approaches focusing on metric learning, prototype learning, and multi-feature integration. Classic works include deep meta-metric learning frameworks such as RS-MetaNet and SPNet [38], [39], which organize task-level learning to improve generalization to novel scene classes with only a few labeled examples. More recent research has explored contrastive and attention-based strategies to handle large intra-class variation and complex backgrounds in remote sensing scenes [40]. Emerging methods further incorporate generative techniques to rectify class prototypes under sample scarcity [41], while vision–language adaptation with prompt learning has been explored to improve cross-domain generalization in few-shot remote sensing scene classification [42].

Recent studies have also paid increasing attention to cross-domain few-shot learning in hyperspectral remote sensing. Yan et al. [43] proposed a semantic-anchor-enhanced prototype learning framework, in which trainable semantic anchors are introduced to guide prototype construction and a lightweight contrastive objective is used to align class prototypes across domains. Zhang et al. [44] developed an orthogonal feature disentanglement network that decomposes representations into domain-shared and domain-specific subspaces, aiming to retain transferable discriminative semantics while suppressing domain-specific interference. Li et al. [45] further introduced a dual prompt-driven cross-modal fusion framework that combines static and learnable prompts with cross-modal representation learning for few-shot hyperspectral image classification. These studies highlight the importance of prototype adaptation, domain-invariant representation learning, and cross-modal semantic guidance for improving few-shot generalization under heterogeneous remote sensing distributions. Different from the above methods, which mainly focus on improving feature representations, prototype construction, domain disentanglement, or multimodal guidance, our work investigates the reconstruction mechanism itself. Specifically, URN targets the representation and reconstruction biases introduced by one-stage support–query projection and addresses them through a cyclic forward–inverse reconstruction framework. Therefore, our method is complementary to existing representation-oriented and cross-domain few-shot learning strategies. Beyond few-shot classification, recent studies [46]–[48] have also explored broader learning-based applications in remote sensing and intelligent sensing. These studies reflect the growing use of intelligent learning techniques in remote sensing and geospatial applications. In this work, we focus specifically on few-shot fine-grained classification under limited labeled data.

C. Reconstruction-Based Methods in Few-Shot Classification

Reconstruction-based methods [6], [24]–[31] represent a prominent line of work in few-shot fine-grained image classification. These approaches typically adopt the following pipeline: *(i) Projection*: support and query features are projected into one another's feature spaces; *(ii) Reconstruction*: new features are reconstructed from the projected representations; *(iii) Metric*

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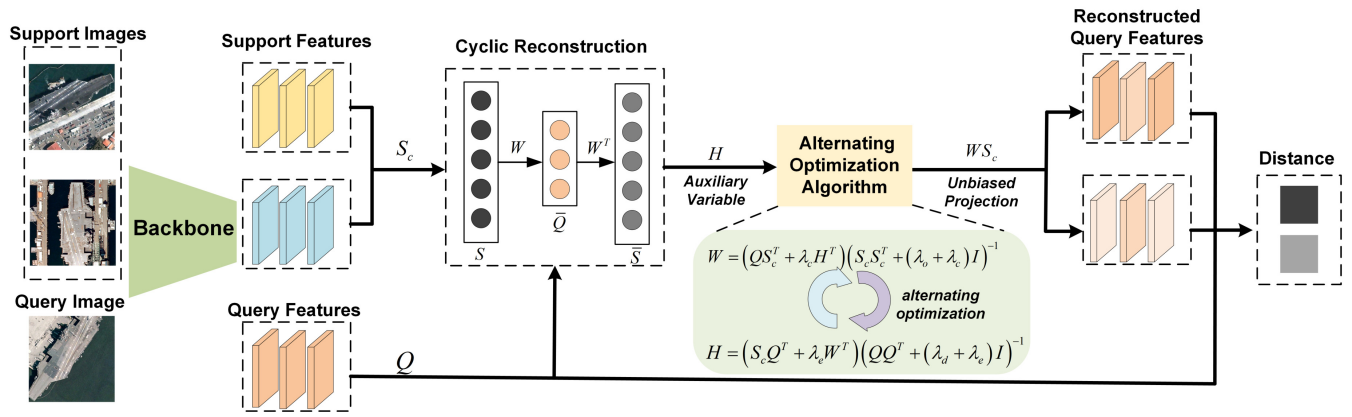


Fig. 2. Diagram of the URN architecture. First, support images and query images are processed through the backbone to obtain support features and query features. Next, a cyclic reconstruction objective function is formulated, and the unbiased projection matrix is obtained through an alternating optimization algorithm. Then, the unbiased projection matrix is applied to the support features to derive reconstructed query features. Finally, the reconstruction error is used as the distance measure between the support features and query features.

computation: similarity is computed based on reconstruction errors, where lower errors indicate higher similarity.

Most existing approaches follow a unified *one-stage reconstruction* paradigm, where each projection is conducted in a single pass and optimized independently. For example, FRN [6] employs ridge regression to learn a support-to-query projection, using the residuals for classification—forming a single-path reconstruction. BiFRN [24] and SRN [27] further introduce a reverse query-to-support mapping, resulting in a two-path structure; however, both projections are learned separately and lack interaction. LCCRN [25] decomposes features into global and local parts, leading to four independent cross-type reconstructions whose residuals are fused. EFRN [26] replaces the L2 norm in FRN with an L2,1-norm penalty to promote group sparsity and improve robustness against noise.

While effective, these one-stage methods suffer from two fundamental limitations:

(1) *Representation bias.* Single-step projections are often low-rank or ill-conditioned in few-shot settings, leading to the loss of fine-grained discriminative cues that cannot be recovered by adding reconstruction paths or residuals.

(2) *Reconstruction bias.* Due to distributional shift between base and novel categories, the encoder often yields novel-class embeddings with large intra-class variance and overlapping inter-class distributions. This structural misalignment causes the reconstruction error to be unreliable—same-class pairs may yield higher errors than different-class ones—thus undermining the metric assumption and impairing discrimination.

To overcome these issues, we propose a two-stage reconstruction framework with an inverse reconstruction constraint at the first time. By reprojecting reconstructed query features back into the support space, this two-stage design enhances alignment, preserves fine-grained semantics, and mitigates both biases jointly.

III. Methodology

A. Revisit One-Stage Reconstruction Network

For the n-way k-shot learning, each episode contains n classes and each class has k support images. For each class $c \in C$, the support features are denoted as $\boldsymbol{S}_c \in \mathbb{R}^{kr \times d}$, and the query features as $\boldsymbol{Q} \in \mathbb{R}^{r \times d}$, where $r = H \times W$ represents the spatial resolution of the feature map, with H and W denoting its height and width, respectively. Metric learning-based methods aim to learn a distance metric function that measures the distance between support features and query features. To learn an accurate distance metric, the pioneering one-stage reconstruction method—Feature Reconstruction Network (FRN) [6] linearly projects the support features $\boldsymbol{S}_c \in \mathbb{R}^{kr \times d}$ into the query feature space to obtain reconstructed query features $\bar{\boldsymbol{Q}} \in \mathbb{R}^{r \times d}$, and the reconstruction error between the reconstructed query features and the real query features $\boldsymbol{Q} \in \mathbb{R}^{r \times d}$ is used as the distance metric. The projection matrix $\boldsymbol{W} \in \mathbb{R}^{r \times kr}$ is obtained by optimizing the following objective function:

$$\min_{\boldsymbol{W}} \|\boldsymbol{Q} - \boldsymbol{W}\boldsymbol{S}_c\|_F^2 + \lambda_o \|\boldsymbol{W}\|_F^2 \tag{1}$$

where $\|\cdot\|_F$ denotes the Frobenius norm, λ_o represents the regularization parameter. By taking the derivative of the above equation, the optimal $\boldsymbol{W}$ can be obtained:

$$\boldsymbol{W} = \boldsymbol{Q}\boldsymbol{S}_c^T\left(\boldsymbol{S}_c\boldsymbol{S}_c^T + \lambda_o \boldsymbol{I}\right)^{-1} \tag{2}$$

Based on the optimal $\boldsymbol{W}$ and $\boldsymbol{S}_c$,The reconstructed query features $\bar{\boldsymbol{Q}}$ can be further calculated as,

$$\bar{\boldsymbol{Q}} = \boldsymbol{Q}\boldsymbol{S}_c^T\left(\boldsymbol{S}_c\boldsymbol{S}_c^T + \lambda_o \boldsymbol{I}\right)^{-1}\boldsymbol{S}_c \tag{3}$$

With the reconstructed query features $\bar{\boldsymbol{Q}}$ and real query features $\boldsymbol{Q}$, FRN adopts the mean square error as the reconstruction error,

$$\langle \boldsymbol{Q}, \bar{\boldsymbol{Q}} \rangle = \frac{1}{r}\left\|\boldsymbol{Q} - \bar{\boldsymbol{Q}}\right\|_F^2 \tag{4}$$

Based on the reconstruction error, the ultimate category prediction is determined by the following formulation:

$$P\left(y_q = c \,|\, x_q\right) = \frac{e^{\left(-\gamma\langle \boldsymbol{Q}, \bar{\boldsymbol{Q}}_c \rangle\right)}}{\sum_{i \in C} e^{\left(-\gamma\langle \boldsymbol{Q}, \bar{\boldsymbol{Q}}_i \rangle\right)}} \tag{5}$$

where γ denotes the learnable temperature coefficient, x_q denotes the query image, y_q denotes the class label of the

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query image x_q, $\bar{\boldsymbol{Q}}_c$ denotes the reconstructed query features by support images with class c.

Although FRN serves as the cornerstone of reconstruction-based method, it adheres to a one-stage reconstruction paradigm—support features are projected into the query space through a single-step linear mapping.

Several subsequent methods [24]–[27] build upon FRN by introducing additional reconstruction paths, regularization terms, or multi-granular feature interactions to enhance reconstruction quality. However, despite these enhancements, they fundamentally retain the same one-stage paradigm: each projection is independently learned in a forward-only manner, without any projection constraints.

B. Biases in One-Stage Reconstruction Network

While these one-stage approaches achieve promising results, they suffer from two intrinsic limitations—*representation bias* and *reconstruction bias*—that hinder the faithful transfer of fine-grained information under the few-shot regime. We now provide a theoretical analysis of these biases.

Representation Bias: One-stage reconstruction methods learn a projection matrix $\boldsymbol{W} \in \mathbb{R}^{r\times kr}$ to reconstruct the query representation as

$$\bar{\boldsymbol{Q}} = \boldsymbol{W}\boldsymbol{S}_c. \tag{6}$$

When $kr > r$, the projection $\boldsymbol{W}$ maps the kr support descriptors into an r-dimensional descriptor space and therefore cannot be injective over the entire input space. Specifically,

$$\operatorname{rank}(\boldsymbol{W}) \le r < kr, \tag{7}$$

which implies a non-trivial null space,

$$\dim \operatorname{Null}(\boldsymbol{W}) \ge kr - r. \tag{8}$$

For the actual support representation $\boldsymbol{S}_c$, the rank of the reconstructed representation satisfies

$$\operatorname{rank}(\boldsymbol{W}\boldsymbol{S}_c) \le \min\left(\operatorname{rank}(\boldsymbol{W}), \operatorname{rank}(\boldsymbol{S}_c)\right) \le \min\left(r, \operatorname{rank}(\boldsymbol{S}_c)\right). \tag{9}$$

Therefore, rank reduction is unavoidable when $\operatorname{rank}(\boldsymbol{S}_c) > r$. When $\operatorname{rank}(\boldsymbol{S}_c) \le r$, however, rank preservation is theoretically possible. This does not imply that all discriminative information is faithfully preserved, because severe attenuation may still occur when the support feature matrix is ill-conditioned.

To further analyze this case, recall that the one-stage ridge reconstruction in Eq. (3) can be written as

$$\bar{\boldsymbol{Q}} = \boldsymbol{Q}\boldsymbol{P}_S, \tag{10}$$

where

$$\boldsymbol{P}_S = \boldsymbol{S}_c^T\left(\boldsymbol{S}_c\boldsymbol{S}_c^T + \lambda_o\boldsymbol{I}\right)^{-1}\boldsymbol{S}_c \tag{11}$$

denotes the effective reconstruction operator induced by the support features. Let the singular value decomposition of $\boldsymbol{S}_c$ be

$$\boldsymbol{S}_c = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T, \tag{12}$$

with nonzero singular values $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_m > 0$. Then $\boldsymbol{P}_S$ can be expressed as

$$\boldsymbol{P}_S = \boldsymbol{V}\operatorname{diag}\left(\frac{\sigma_1^2}{\sigma_1^2+\lambda_o}, \ldots, \frac{\sigma_m^2}{\sigma_m^2+\lambda_o}\right)\boldsymbol{V}^T. \tag{13}$$

Thus, the component along the i-th singular direction is scaled by

$$\alpha_i = \frac{\sigma_i^2}{\sigma_i^2+\lambda_o}, \qquad 0 < \alpha_i < 1. \tag{14}$$

This spectral form reveals that rank preservation alone is insufficient to guarantee information preservation. For an ill-conditioned support matrix, defined by

$$\kappa(\boldsymbol{S}_c) = \frac{\sigma_{\max}}{\sigma_{\min}}, \tag{15}$$

a large condition number indicates the existence of a very small $\sigma_{\min}$. The weakest retained direction is therefore scaled by

$$\alpha_{\min} = \frac{\sigma_{\min}^2}{\sigma_{\min}^2+\lambda_o} = \frac{\sigma_{\max}^2}{\sigma_{\max}^2+\lambda_o\kappa^2(\boldsymbol{S}_c)}. \tag{16}$$

For a fixed $\sigma_{\max}$ and λ_o, $\alpha_{\min}$ decreases as the condition number increases and approaches zero for a severely ill-conditioned matrix. Consequently, a direction may remain mathematically present, i.e., without causing any rank reduction, while its contribution to the reconstructed representation becomes negligible. This effect is particularly important for fine-grained recognition, where subtle discriminative cues may reside in such low-energy feature directions.

A simple numerical example illustrates this phenomenon. Consider

$$\boldsymbol{S}_c = \operatorname{diag}(1, \epsilon), \qquad \lambda_o = 10^{-2}. \tag{17}$$

Although $\boldsymbol{S}_c$ is full rank for any $\epsilon > 0$, its effective reconstruction operator is

$$\boldsymbol{P}_S = \operatorname{diag}\left(\frac{1}{1.01}, \frac{\epsilon^2}{\epsilon^2+0.01}\right). \tag{18}$$

When $\epsilon = 0.1$, the condition number is 10, and the second direction is scaled by 0.5. When $\epsilon = 0.01$, the matrix remains full rank but its condition number increases to 100, while the corresponding scaling factor decreases to approximately 0.0099. Hence, more than 99% of the information magnitude along this direction is suppressed even though no rank loss occurs.

Therefore, representation bias can arise from two closely related mechanisms. First, dimensional compression or rank deficiency may completely eliminate support components through the null space of the projection. Second, even when the relevant rank is preserved, ill-conditioning can strongly attenuate components associated with small singular values. We refer to such projection-induced loss or suppression of potentially discriminative information as *representation bias*.

Reconstruction Bias: Consider the meta-testing phase, where the projection matrix $\boldsymbol{W}_{\text{novel}}$ is re-estimated from novel-class support-query pairs $(\boldsymbol{S}_{\text{novel}}, \boldsymbol{Q}_{\text{novel}})$. Although $\boldsymbol{W}_{\text{novel}}$ is optimized within each episode, both $\boldsymbol{S}_{\text{novel}}$ and $\boldsymbol{Q}_{\text{novel}}$ are embedded by a backbone encoder $\phi(\cdot)$ pretrained on base classes. Let $P_{\text{emb}}^{(c)}$ denote the embedding distribution for class c.

For base classes, the encoder generally preserves the metric-learning property:

$$\mathbb{E}\left[d_{\text{intra}}^{\text{base}}\right] \ll \mathbb{E}\left[d_{\text{inter}}^{\text{base}}\right], \tag{19}$$

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where d_{intra} and d_{inter} denote the intra-class and inter-class distances in the embedding space.

However, for novel classes drawn from a shifted distribution, the encoder may induce enlarged intra-class variance:

$$\operatorname{tr}(\Sigma_{\text{novel}}) \gg \operatorname{tr}(\Sigma_{\text{base}}), \tag{20}$$

where Σ_{novel} is the covariance of $P_{\text{emb}}^{(c_{\text{novel}})}$. In extreme cases, the intra-class distance for novel categories can exceed the inter-class separation:

$$\mathbb{E}\big[d_{\text{intra}}^{\text{novel}}\big] > \mathbb{E}\big[d_{\text{inter}}^{\text{novel}}\big], \tag{21}$$

breaking the separation assumption.

Since the projection matrix $\boldsymbol{W}_{\text{novel}}$ is learned from such misaligned embeddings, the reconstruction error

$$\mathcal{E}_{\text{rec}}(q,s) = \|\phi(q) - \boldsymbol{W}_{\text{novel}}\phi(s)\|_2^2 \tag{22}$$

will inherit this distortion. Consequently, the abnormal condition

$$\mathbb{E}\left[\mathcal{E}_{\text{rec}}^{\text{intra}}\right] \gtrsim \mathbb{E}\left[\mathcal{E}_{\text{rec}}^{\text{inter}}\right] \tag{23}$$

emerges, meaning that same-class pairs may incur equal or even larger reconstruction errors than different-class pairs.

We define this problem as *reconstruction bias*, caused by distributional shift between base and novel classes that inflates intra-class distances and violates the fundamental metric-learning assumption. This bias severely degrades fine-grained discrimination in few-shot scenarios.

Representation bias and reconstruction bias collectively hinder the effectiveness of one-stage reconstruction networks. To mitigate these limitations, we propose the *Unbiased Reconstruction Network* (URN), a novel two-stage reconstruction framework, as described in the following section.

C. Unbiased Reconstruction Network

To overcome the challenges of representation and reconstruction biases inherent in one-stage reconstruction methods, we propose the first two-stage reconstruction framework, termed the Unbiased Reconstruction Network (URN). The overall architecture of URN is illustrated in Figure 2. Specifically, an inverse reconstruction constraint is introduced, where after projecting $\boldsymbol{S}_c$ into the query feature space to obtain the reconstructed query features $\bar{\boldsymbol{Q}}$, $\bar{\boldsymbol{Q}}$ is further projected back into the support feature space to obtain the reconstructed support features $\bar{\boldsymbol{S}}_c$. To implement the inverse reconstruction constraint, a novel cyclic reconstruction objective function is proposed, which is defined as follows:

$$\begin{cases} \min\limits_{\boldsymbol{W}} \left\|\boldsymbol{S}_c - \boldsymbol{W}^T\boldsymbol{W}\boldsymbol{S}_c\right\|_F^2 + \lambda_o \|\boldsymbol{W}\|_F^2 \\ \text{s.t.} \quad \boldsymbol{W}\boldsymbol{S}_c \approx \boldsymbol{Q} \end{cases} \tag{24}$$

where $\boldsymbol{W}$ is the forward projection matrix (support-to-query), and $\boldsymbol{W}^T$ serves as a relaxed backward projection (query-to-support). From Eq. (24), it can be seen that the proposed cyclic reconstruction objective imposes two layers of constraints: The first constraint ensures that the reconstructed query features $\bar{\boldsymbol{Q}}$ are as consistent with the query features $\boldsymbol{Q}$ as possible, which aligns with the approach of FRN. The second constraint requires that the reconstructed support features $\bar{\boldsymbol{S}}_c$ obtained by inverse projection of $\bar{\boldsymbol{Q}}$ are as consistent with the original support features $\boldsymbol{S}_c$ as possible, which represents the proposed inverse reconstruction constraint. We now theoretically explain how this objective mitigates both types of bias.

(1) Mitigating Representation Bias: In one-stage reconstruction, the projection matrix $\boldsymbol{W} \in \mathbb{R}^{r\times kr}$ maps the support representation $\boldsymbol{S}_c \in \mathbb{R}^{kr\times d}$ into a lower-dimensional descriptor space. As analyzed in Section III-B, representation bias may arise from two related mechanisms. First, when $kr > r$, $\boldsymbol{W}$ is necessarily non-injective over the entire input space and has a non-trivial null space. Therefore, support components aligned with the null space may be completely discarded during projection. Moreover, when $\operatorname{rank}(\boldsymbol{S}_c) > r$, rank reduction of the projected representation is unavoidable. Second, even when the effective rank is preserved, an ill-conditioned support representation may contain directions associated with very small singular values. Under ridge-regression reconstruction, these directions are strongly attenuated by the shrinkage factor

$$\alpha_i = \frac{\sigma_i^2}{\sigma_i^2 + \lambda_o}, \tag{25}$$

and may therefore contribute only marginally to the reconstructed representation.

To alleviate both effects, URN introduces a cyclic reconstruction constraint that requires the support representation to remain recoverable after the forward and inverse mappings. In its conceptual form, this constraint can be written as

$$\boldsymbol{S}_c \approx \boldsymbol{W}^T\boldsymbol{W}\boldsymbol{S}_c. \tag{26}$$

Rather than requiring $\boldsymbol{W}^T\boldsymbol{W} \approx \boldsymbol{I}_{kr}$ over the entire input space, which is generally impossible when $r < kr$, this constraint encourages $\boldsymbol{W}^T\boldsymbol{W}$ to behave approximately as an identity mapping on the recoverable subspace occupied by $\boldsymbol{S}_c$.

This requirement imposes an additional penalty on projection directions that excessively discard or attenuate information. To see this more clearly, consider a support component $\boldsymbol{s}_i$ associated with a recoverable direction. If the forward projection strongly suppresses this component, i.e.,

$$\|\boldsymbol{W}\boldsymbol{s}_i\|_2 \ll \|\boldsymbol{s}_i\|_2, \tag{27}$$

then the inverse reconstruction cannot faithfully recover the original component, leading to a large cyclic reconstruction residual,

$$\left\|\boldsymbol{s}_i - \boldsymbol{W}^T\boldsymbol{W}\boldsymbol{s}_i\right\|_2^2. \tag{28}$$

Consequently, such strongly attenuated solutions are explicitly penalized during optimization. This mechanism is particularly relevant for ill-conditioned feature matrices, where subtle discriminative cues may lie along small-singular-value directions and would otherwise be severely suppressed by one-stage ridge reconstruction.

More generally, the implemented URN introduces an auxiliary inverse mapping $\boldsymbol{H}$ and couples it with the forward mapping $\boldsymbol{W}$ through consistency regularization. The resulting optimization encourages

$$\boldsymbol{H}\boldsymbol{W}\boldsymbol{S}_c \approx \boldsymbol{S}_c, \tag{29}$$

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while simultaneously enforcing

$$\boldsymbol{W}\boldsymbol{S}_c \approx \boldsymbol{Q}. \tag{30}$$

Therefore, the learned forward projection is not determined solely by its ability to approximate the query representation, but is also constrained by whether the retained information can be recovered through the inverse mapping.

From this perspective, the cyclic reconstruction constraint does not remove the fundamental dimensional limitation of $\boldsymbol{W}$, nor can it guarantee perfect preservation of directions that are inherently unrecoverable. Instead, it reduces the tendency of one-stage optimization to select degenerate mappings that unnecessarily discard or strongly attenuate recoverable support information. By explicitly favoring projections with improved recoverability, URN preserves more fine-grained discriminative structure and thereby mitigates representation bias.

(2) Mitigating Reconstruction Bias: Reconstruction bias arises when the support and query representations of novel classes are not well aligned in feature space. Although the projection matrix is re-estimated during meta-testing, both support and query features are extracted by an encoder learned from base classes. Consequently, distribution shift may lead to a mismatch between the subspaces spanned by novel support and query representations, making one-way reconstruction less reliable.

This effect can be characterized more formally from a subspace projection perspective. Consider the unregularized forward reconstruction

$$\min_{\boldsymbol{W}} \|\boldsymbol{Q} - \boldsymbol{W}\boldsymbol{S}_c\|_F^2 . \tag{31}$$

Let

$$\boldsymbol{P}_S = \boldsymbol{S}_c^T \left(\boldsymbol{S}_c\boldsymbol{S}_c^T\right)^\dagger \boldsymbol{S}_c \tag{32}$$

denote the orthogonal projector onto the row subspace spanned by $\boldsymbol{S}_c$. The optimal one-stage reconstruction can then be written as

$$\bar{\boldsymbol{Q}} = \boldsymbol{Q}\boldsymbol{P}_S, \tag{33}$$

and the corresponding minimum reconstruction error becomes

$$\left\|\boldsymbol{Q} - \bar{\boldsymbol{Q}}\right\|_F^2 = \|\boldsymbol{Q}\left(\boldsymbol{I} - \boldsymbol{P}_S\right)\|_F^2 . \tag{34}$$

Therefore, the forward reconstruction error is determined by the component of the query representation that cannot be expressed by the support subspace. When the support–query discrepancy increases on novel classes, this unrepresentable component may become larger, resulting in reconstruction bias.

URN alleviates this asymmetric limitation by additionally introducing inverse reconstruction. Let

$$\boldsymbol{P}_Q = \boldsymbol{Q}^T \left(\boldsymbol{Q}\boldsymbol{Q}^T\right)^\dagger \boldsymbol{Q} \tag{35}$$

denote the projector onto the query row subspace. The corresponding inverse reconstruction error can be expressed as

$$\|\boldsymbol{S}_c\left(\boldsymbol{I} - \boldsymbol{P}_Q\right)\|_F^2 . \tag{36}$$

Accordingly, the forward and inverse reconstruction terms jointly characterize a bidirectional subspace discrepancy:

$$\mathcal{D}_{\text{cyc}} = \|\boldsymbol{Q}\left(\boldsymbol{I} - \boldsymbol{P}_S\right)\|_F^2 + \|\boldsymbol{S}_c\left(\boldsymbol{I} - \boldsymbol{P}_Q\right)\|_F^2 . \tag{37}$$

Unlike one-stage reconstruction, which only requires the query representation to be approximated from the support representation, the cyclic formulation encourages mutual representability between the support and query subspaces. Therefore, URN reduces reconstruction asymmetry and encourages more consistent support–query reconstruction, thereby mitigating reconstruction bias.

D. Challenges in Optimization

To optimize the objective function in Eq. (24), we can reformulate it as Eq. (38):

$$\min_{\boldsymbol{W}} \left\|\boldsymbol{S}_c - \boldsymbol{W}^T\boldsymbol{Q}\right\|_F^2 + \lambda_o \|\boldsymbol{W}\|_F^2 + \lambda_f \|\boldsymbol{Q} - \boldsymbol{W}\boldsymbol{S}_c\|_F^2 \tag{38}$$

where λ_f denotes the weighting coefficient that controls the importance of the constraint $\boldsymbol{W}\boldsymbol{S}_c \approx \boldsymbol{Q}$.

To facilitate the differentiation with respect to $\boldsymbol{W}$, Eq. (38) can be further transformed into:

$$\min_{\boldsymbol{W}} \left\|\boldsymbol{S}_c^T - \boldsymbol{Q}^T\boldsymbol{W}\right\|_F^2 + \lambda_o \|\boldsymbol{W}\|_F^2 + \lambda_f \|\boldsymbol{Q} - \boldsymbol{W}\boldsymbol{S}_c\|_F^2 \tag{39}$$

Taking the derivative of Eq. (39) yields:

$$\left(\boldsymbol{Q}\boldsymbol{Q}^T + \lambda_o\boldsymbol{I}\right)\boldsymbol{W} + \lambda_f\boldsymbol{W}\boldsymbol{S}_c\boldsymbol{S}_c^T = (1+\lambda_f)\,\boldsymbol{Q}\boldsymbol{S}_c^T \tag{40}$$

Assuming $\boldsymbol{A} = \left(\boldsymbol{Q}\boldsymbol{Q}^T + \lambda_o\boldsymbol{I}\right)$, $\boldsymbol{B} = \lambda_f\boldsymbol{S}_c\boldsymbol{S}_c^T$, $\boldsymbol{C} = (1+\lambda_f)\,\boldsymbol{Q}\boldsymbol{S}_c^T$, Eq. (40) can be written in the following form:

$$\boldsymbol{A}\boldsymbol{W} + \boldsymbol{W}\boldsymbol{B} = \boldsymbol{C} \tag{41}$$

Eq. (41) is a classic Sylvester equation, for which no closed-form solution exists. While classical numerical methods such as the Bartels-Stewart algorithm [32] are designed to solve such equations, they are not suitable for our learning framework due to several critical limitations:

First, the Bartels-Stewart algorithm relies on computing the real Schur decompositions of $\boldsymbol{A}$ and $\boldsymbol{B}$, which has a time complexity of $\mathcal{O}(n^3)$ and memory complexity of $\mathcal{O}(n^2)$ for $n \times n$ matrices. As both $\boldsymbol{Q}\boldsymbol{Q}^T$ and $\boldsymbol{S}_c\boldsymbol{S}_c^T$ can be large in high-dimensional few-shot feature spaces (especially when r is large), the memory and computation costs quickly become prohibitive.

Second, Schur decomposition and the subsequent back-substitution steps are inherently sequential and exhibit poor parallelism. This makes the Bartels-Stewart method difficult to implement efficiently on modern hardware accelerators (e.g., GPUs), which are optimized for batched and parallelized matrix operations. Consequently, integrating it into end-to-end deep learning pipelines substantially slows down training and hinders scalability.

Third, and most crucially, we observe in practice that using Bartels-Stewart within our training loop often yields unstable numerical results. Since neural network gradients are highly sensitive to numerical precision, solving Eq. (41) via Schur-based methods leads to ill-conditioned matrices or even NaN gradients during backpropagation. This can cause the network to diverge or fail to converge altogether.

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E. Auxiliary Variable for Efficient Optimization

To address the challenge of quickly and stably solving the unbiased projection matrix $\boldsymbol{W}$, we introduce an auxiliary matrix variable $\boldsymbol{H}$ and reformulate the original objective function as follows:

$$\begin{cases} \min\limits_{\boldsymbol{W},\boldsymbol{H}} \|\boldsymbol{Q}-\boldsymbol{W}\boldsymbol{S}_c\|_F^2+\lambda_o\|\boldsymbol{W}\|_F^2+\lambda_c\left\|\boldsymbol{W}-\boldsymbol{H}^T\right\|_F^2 \\ \text{s.t.} \quad \boldsymbol{H}\boldsymbol{Q}\approx\boldsymbol{S}_c \end{cases} \tag{42}$$

where λ_c is a regularization hyperparameter that controls the degree of consistency between $\boldsymbol{W}$ and the transpose of $\boldsymbol{H}$.

The motivation for introducing the auxiliary variable $\boldsymbol{H}$ stems from variable splitting techniques commonly employed in constrained optimization. The core idea is to decouple a difficult bilinear or nonlinear objective into multiple tractable subproblems by introducing a surrogate variable that approximates part of the transformation.

In our case, the original cyclic reconstruction objective involves both forward and backward projections through $\boldsymbol{W}$ and $\boldsymbol{W}^T$, which makes direct optimization difficult due to the presence of a Sylvester-type coupling:

$$\boldsymbol{A}\boldsymbol{W}+\boldsymbol{W}\boldsymbol{B}=\boldsymbol{C} \tag{43}$$

where no closed-form solution exists and numerical solvers such as the Bartels-Stewart algorithm are computationally expensive and unstable during training.

By replacing $\boldsymbol{W}^T$ with a decoupled variable $\boldsymbol{H}$ and enforcing the consistency constraint $\boldsymbol{W}\approx\boldsymbol{H}^T$, the optimization can be split into two subproblems over $\boldsymbol{W}$ and $\boldsymbol{H}$, both of which admit closed-form solutions via ridge regression. Specifically:

- When fixing $\boldsymbol{H}$, the subproblem with respect to $\boldsymbol{W}$ becomes a standard ℓ_2-regularized least squares problem.
- When fixing $\boldsymbol{W}$, the subproblem with respect to $\boldsymbol{H}$ reduces to a pseudo-inverse-based projection, which is computationally efficient.

This reformulation is theoretically justified by the augmented Lagrangian principle in convex optimization, where the consistency regularization $\left\|\boldsymbol{W}-\boldsymbol{H}^T\right\|_F^2$ can be interpreted as a soft constraint enforcing the equivalence of two latent mappings. Moreover, it preserves the expressiveness of the original objective while substantially improving numerical stability and convergence speed during training.

Therefore, the auxiliary variable $\boldsymbol{H}$ enables an efficient alternating optimization framework, bypasses the limitations of Sylvester equation solvers, and ensures that the learned projection remains structurally aligned across both forward and backward mappings. Detailed update rules are described in the next section.

To solve Eq. (42), we first incorporate the constraint $\boldsymbol{H}\boldsymbol{Q}\approx\boldsymbol{S}_c$ into the objective function as two penalty terms:

$$\begin{aligned} \min_{\boldsymbol{W},\boldsymbol{H}} &\|\boldsymbol{W}\boldsymbol{S}_c-\boldsymbol{Q}\|_F^2+\lambda_o\|\boldsymbol{W}\|_F^2+\lambda_c\left\|\boldsymbol{W}-\boldsymbol{H}^T\right\|_F^2 \\ &+\lambda_b\|\boldsymbol{H}\boldsymbol{Q}-\boldsymbol{S}_c\|_F^2+\lambda_d\|\boldsymbol{H}\|_F^2 \end{aligned} \tag{44}$$

where the $\|\boldsymbol{H}\|_F^2$ is introduced to prevent overfitting of $\boldsymbol{H}$, λ_b and λ_d respectively denote the weighting coefficient that controls the importance of $\|\boldsymbol{H}\boldsymbol{Q}-\boldsymbol{S}_c\|_F^2$ and $\|\boldsymbol{H}\|_F^2$.

The objective function in Eq. (44) is quadratic in form, making it a convex function with a global optimum. The alternating optimization algorithm is employed to obtain the optimal $\boldsymbol{W}$ by alternatingly solving the following subproblems:

$$\begin{cases} \min\limits_{\boldsymbol{W}} \|\boldsymbol{W}\boldsymbol{S}_c-\boldsymbol{Q}\|_F^2+\lambda_o\|\boldsymbol{W}\|_F^2+\lambda_c\left\|\boldsymbol{W}-\boldsymbol{H}^T\right\|_F^2 \\ \min\limits_{\boldsymbol{H}} \|\boldsymbol{H}\boldsymbol{Q}-\boldsymbol{S}_c\|_F^2+\lambda_d\|\boldsymbol{H}\|_F^2+\lambda_e\left\|\boldsymbol{W}-\boldsymbol{H}^T\right\|_F^2 \end{cases} \tag{45}$$

We detail the solution to each subproblem as follows: **Subproblem $\boldsymbol{W}$**: We take the derivative of the objective function of the subproblem with respect to $\boldsymbol{W}$ and set it to zero. After rearranging, we can obtain:

$$\boldsymbol{W}\left(\boldsymbol{S}_c\boldsymbol{S}_c^T+(\lambda_o+\lambda_c)\boldsymbol{I}\right)=\boldsymbol{Q}\boldsymbol{S}_c^T+\lambda_c\boldsymbol{H}^T \tag{46}$$

Thus, the closed-form solution for $\boldsymbol{W}$ is given by:

$$\boldsymbol{W}=\left(\boldsymbol{Q}\boldsymbol{S}_c^T+\lambda_c\boldsymbol{H}^T\right)\left(\boldsymbol{S}_c\boldsymbol{S}_c^T+(\lambda_o+\lambda_c)\boldsymbol{I}\right)^{-1} \tag{47}$$

Subproblem $\boldsymbol{H}$: Next, we solve the second subproblem in Eq. (45). Similarly, by taking the derivative of the objective function with respect to $\boldsymbol{H}$ and setting it to zero, we can obtain:

$$\boldsymbol{H}\left(\boldsymbol{Q}\boldsymbol{Q}^T+(\lambda_d+\lambda_e)\boldsymbol{I}\right)=\left(\boldsymbol{S}_c\boldsymbol{Q}^T+\lambda_e\boldsymbol{W}^T\right) \tag{48}$$

Then the closed-form solution for $\boldsymbol{H}$ can be expressed as:

$$\boldsymbol{H}=\left(\boldsymbol{S}_c\boldsymbol{Q}^T+\lambda_e\boldsymbol{W}^T\right)\left(\boldsymbol{Q}\boldsymbol{Q}^T+(\lambda_d+\lambda_e)\boldsymbol{I}\right)^{-1} \tag{49}$$

To obtain the optimal $\boldsymbol{W}$, we employ the alternating optimization algorithm. First, an initial value is assigned to the auxiliary variable $\boldsymbol{H}$, then $\boldsymbol{W}$ and $\boldsymbol{H}$ are iteratively optimized in an alternating manner. After a certain number of iterations, the unbaised projection matrix $\boldsymbol{W}$ converges to a near-global optimum. Once $\boldsymbol{W}$ is computed, the distance between $\boldsymbol{S}_c$ and $\boldsymbol{Q}$ can be determined. The final algorithm flowchart can be summarized as Algorithm 1:

Algorithm 1 Alternating Optimization Algorithm for URN

Require: Support feature $\mathbf{S}_c$, query feature $\mathbf{Q}$, number of iterations N, initial value $\boldsymbol{H}^0$

Ensure: Converged projection matrix $\mathbf{W}$, distance d_c between $\mathbf{S}_c$ and $\mathbf{Q}$

$\boldsymbol{H}=\boldsymbol{H}^0$
for $i=1:N$ **do**
  $\boldsymbol{W}=\left(\boldsymbol{Q}\boldsymbol{S}_c^T+\lambda_c\boldsymbol{H}^T\right)\left(\boldsymbol{S}_c\boldsymbol{S}_c^T+(\lambda_o+\lambda_c)\boldsymbol{I}\right)^{-1}$
  $\boldsymbol{H}=\left(\boldsymbol{S}_c\boldsymbol{Q}^T+\lambda_e\boldsymbol{W}^T\right)\left(\boldsymbol{Q}\boldsymbol{Q}^T+(\lambda_d+\lambda_e)\boldsymbol{I}\right)^{-1}$
end for
$\bar{\boldsymbol{Q}}_c=\mathbf{W}\mathbf{S}_c$
$d_c=\left\langle\boldsymbol{Q},\bar{\boldsymbol{Q}}\right\rangle=\frac{1}{r}\left\|\boldsymbol{Q}-\bar{\boldsymbol{Q}}\right\|^2$

Testing Phase: Based on the distance d_c, the category probability of the query image x_q can be computed using Eq. (5). The category with the highest probability is then assigned as the predicted label for x_q.

Complexity Analysis: We first provide a conservative complexity analysis by treating the feature matrices as dense matrices. Given $\boldsymbol{Q}\in\mathbb{R}^{r\times d}$ and $\boldsymbol{S}_c\in\mathbb{R}^{kr\times d}$, where r denotes the number of spatial descriptors, the update of $\boldsymbol{W}$

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requires computing $\boldsymbol{Q}\boldsymbol{S}_c^\top$ and $\boldsymbol{S}_c\boldsymbol{S}_c^\top$. Their computational complexities are $\mathcal{O}(kdr^2)$ and $\mathcal{O}(dk^2r^2)$, respectively. Solving the corresponding $(kr)\times(kr)$ linear system requires $\mathcal{O}(k^3r^3)$ operations in the standard dense case. Therefore, the dominant complexity of the $\boldsymbol{W}$ subproblem is

$$\mathcal{O}(dk^2r^2 + k^3r^3). \tag{50}$$

For the $\boldsymbol{H}$ subproblem, computing $\boldsymbol{S}_c\boldsymbol{Q}^\top$ and $\boldsymbol{Q}\boldsymbol{Q}^\top$ requires $\mathcal{O}(kdr^2)$ and $\mathcal{O}(dr^2)$ operations, respectively, while solving the $r\times r$ linear system requires $\mathcal{O}(r^3)$ operations. Thus, its dominant complexity is

$$\mathcal{O}(kdr^2 + dr^2 + kr^3 + r^3). \tag{51}$$

Accordingly, a conservative upper bound for one alternating iteration is dominated by

$$\mathcal{O}(dk^2r^2 + k^3r^3), \tag{52}$$

and the total complexity for N iterations is

$$\mathcal{O}\left(N(dk^2r^2 + k^3r^3)\right). \tag{53}$$

It is worth noting that the above complexity corresponds to a general dense implementation. The support feature matrix, however, naturally possesses a shot-wise block structure. Specifically, it can be written as

$$\boldsymbol{S}_c = \left[(\boldsymbol{S}_c^{(1)})^T, \ldots, (\boldsymbol{S}_c^{(k)})^T\right]^T, \tag{54}$$

where $\boldsymbol{S}_c^{(i)} \in \mathbb{R}^{r\times d}$ denotes the spatial descriptor matrix of the i-th support image. Consequently, $\boldsymbol{S}_c\boldsymbol{S}_c^T$ consists of $k\times k$ Gram blocks of size $r\times r$.

A useful special case arises when the cross-shot correlations are weak, i.e.,

$$\boldsymbol{S}_c^{(i)}(\boldsymbol{S}_c^{(j)})^T \approx \boldsymbol{0}, \qquad i\neq j. \tag{55}$$

Under this condition, the Gram matrix can be approximated by a block-diagonal form,

$$\boldsymbol{S}_c\boldsymbol{S}_c^T \approx \operatorname{diag}\left(\boldsymbol{S}_c^{(1)}(\boldsymbol{S}_c^{(1)})^T, \ldots, \boldsymbol{S}_c^{(k)}(\boldsymbol{S}_c^{(k)})^T\right). \tag{56}$$

The original $(kr)\times(kr)$ linear system can then be decomposed into k independent $r\times r$ systems. Accordingly, the complexity of the matrix solution is reduced from $\mathcal{O}(k^3r^3)$ to $\mathcal{O}(kr^3)$, while the k blocks can also be processed in parallel. This special case illustrates how the intrinsic shot-wise structure of few-shot feature matrices can potentially be exploited to further reduce the practical computational cost beyond the conservative dense-matrix bound.

Convergence Analysis: We analyze the convergence properties of the proposed alternating optimization scheme used to solve Eq. (45). The optimization objective involves two variables, $\boldsymbol{W}$ and $\boldsymbol{H}$, and can be decomposed into the following two subproblems:

- Fix $\boldsymbol{H}$ and solve a convex quadratic minimization problem with respect to $\boldsymbol{W}$.
- Fix $\boldsymbol{W}$ and solve a convex quadratic minimization problem with respect to $\boldsymbol{H}$.

Both subproblems admit closed-form solutions and are convex with respect to their respective variables. According to the standard alternating minimization principle [49], solving each subproblem exactly ensures that the objective value is monotonically non-increasing during the alternating updates, and the optimization converges to a stationary solution under the adopted setting.

Let $\mathcal{L}^{(t)}$ denote the objective value at the t-th iteration. The alternating updates satisfy

$$\mathcal{L}^{(0)} \geq \mathcal{L}^{(1)} \geq \cdots \geq \mathcal{L}^{(t)} \to \bar{\mathcal{L}}, \tag{57}$$

where $\bar{\mathcal{L}}$ denotes the converged objective value rather than a guaranteed global optimum. Empirically, the optimization reaches a stable region within a few iterations. As verified by the cross-dataset analysis, $N=5$ provides the best or stable performance across the evaluated datasets. Therefore, we set $N=5$ in all experiments to balance computational efficiency and solution stability.

TABLE I
THE EXPERIMENTAL RESULTS FOR **FGSCR** DATASET

Method	Backbone	1-shot	5-shot
RelationNet [3]	Conv-4	42.83 ± 0.16	56.24 ± 0.12
DN4 [4]	Conv-4	34.63 ± 0.60	51.98 ± 0.36
DSN [50]	Conv-4	33.76 ± 0.16	49.39 ± 0.12
RIFRN [30]	Conv-4	53.74 ± 0.17	79.66 ± 0.11
ProtoNet [1]	ResNet-12	36.30 ± 0.17	64.49 ± 0.13
CrossTransformer [7]	ResNet-12	47.49 ± 0.15	64.13 ± 0.14
ReNet [16]	ResNet-12	38.42 ± 0.43	62.97 ± 0.30
FRN [6]	ResNet-12	54.49 ± 0.17	76.80 ± 0.13
TDM [17]	ResNet-12	53.94 ± 0.17	74.38 ± 0.12
FAFRN [28]	ResNet-12	56.91 ± 0.16	79.50 ± 0.11
CWFRN [29]	ResNet-12	56.46 ± 0.16	79.07 ± 0.12
FKFRN [31]	ResNet-12	58.29 ± 0.16	80.73 ± 0.11
Ours	ResNet-12	**60.36 ± 0.16**	**81.93 ± 0.10**

TABLE II
THE EXPERIMENTAL RESULTS FOR **NWPU-RESISC45** DATASET

Method	Backbone	1-shot	5-shot
DLA-MatchNet [51]	Conv-4	68.80 ± 0.70	81.63 ± 0.46
MatchingNet [2]	ResNet-12	61.57 ± 0.49	76.02 ± 0.34
ProtoNet [1]	ResNet-12	64.52 ± 0.48	81.95 ± 0.30
RelationNet [3]	ResNet-12	65.52 ± 0.85	78.38 ± 0.31
FRN [6]	ResNet-12	66.31 ± 0.20	79.67 ± 0.15
DeepEMD [12]	ResNet-12	67.50 ± 0.74	80.54 ± 0.51
SPNet [39]	ResNet-18	67.84 ± 0.87	83.94 ± 0.50
DeepDBC [10]	ResNet-12	68.84 ± 0.44	84.12 ± 0.29
Ours	ResNet-12	**70.23 ± 0.21**	**85.33 ± 0.13**

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: To comprehensively evaluate the effectiveness and generalizability of URN, we conduct extensive experiments on nine diverse datasets covering both remote sensing and natural image domains. Specifically, four remote sensing benchmarks are adopted, including FGSCR [52] for fine-grained ship classification, and NWPU-RESISC45 [53], AID [54], and WHU-RS19 [55] for remote sensing scene classification. We further evaluate URN on four widely used natural fine-grained image classification benchmarks, including CUB [56], Stanford Dogs [57], Stanford Cars [58], and Aircraft [59]. In addition,

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TABLE III
THE EXPERIMENTAL RESULTS FOR THE **AID** DATASET.

Method	Backbone	1-shot	5-shot
MAML [60]	ResNet-12	48.16 ± 0.37	61.13 ± 0.94
Meta-SGD [61]	ResNet-12	50.01 ± 0.93	65.31 ± 0.38
MatchingNet [2]	ResNet-12	42.17 ± 0.78	52.34 ± 0.89
ProtoNet [1]	ResNet-12	49.91 ± 0.47	70.48 ± 0.21
RelationNet [3]	ResNet-12	53.51 ± 0.68	68.65 ± 0.95
DLA-MatchNet [51]	Conv-4	53.41 ± 0.99	70.61 ± 0.36
SCL-MLNet [62]	ResNet-12	59.46 ± 0.96	76.31 ± 0.68
MKN [63]	ResNet-12	57.29 ± 0.59	75.42 ± 0.31
MPCL-Net [64]	ResNet-12	60.61 ± 0.43	76.78 ± 0.08
HiReNet [65]	ResNet-12	59.43 ± 0.66	74.12 ± 0.43
Ours	ResNet-12	**66.36 ± 0.21**	**83.22 ± 0.14**

TABLE IV
THE EXPERIMENTAL RESULTS FOR THE **WHU-RS19** DATASET.

Method	Backbone	1-shot	5-shot
MAML [60]	ResNet-12	50.87 ± 0.23	64.26 ± 0.32
Meta-SGD [61]	ResNet-12	51.78 ± 1.05	65.47 ± 0.65
MatchingNet [2]	ResNet-12	51.25 ± 0.61	54.36 ± 0.38
ProtoNet [1]	ResNet-12	58.17 ± 0.56	80.54 ± 0.42
RelationNet [3]	ResNet-12	61.74 ± 0.51	79.15 ± 0.35
DLA-MatchNet [51]	Conv-4	68.27 ± 1.83	79.89 ± 0.33
SPNet [39]	ResNet-18	81.06 ± 0.60	88.04 ± 0.28
TAE-Net [66]	ResNet-12	73.67 ± 0.74	88.95 ± 0.52
GES-Net [67]	ResNet-12	75.84 ± 0.78	82.37 ± 0.38
Ours	ResNet-12	**84.88 ± 0.13**	**96.24 ± 0.05**

to assess the generalizability of URN beyond fine-grained recognition, we conduct supplementary experiments on the general few-shot classification benchmark mini-ImageNet [2].

2) Implementation Details: Following recent studies [6], [12], [17],the input image size set to 84 × 84. During training, standard data augmentation techniques, including random cropping and horizontal flipping, are applied. To evaluate N-way K-shot classification, each episode includes 16 query images per category. We sample 10,000 random episodes to compute the mean classification accuracy with a 95% confidence interval. In URN, the iterations for the alternating optimization algorithm is set to 5, with the initial value of $\boldsymbol{H}$ initialized as a zero matrix $\boldsymbol{O}$. Additionally, the four regularization parameters, λ_o, λ_c, λ_d, and λ_e, are automatically optimized through network learning.

B. Comparison with State-of-the-Art Methods

1) Few-Shot Fine-Grained Classification: We evaluate URN on eight fine-grained and remote sensing classification benchmarks, including four remote sensing datasets (FGSCR, NWPU-RESISC45, AID, and WHU-RS19) and four natural fine-grained datasets (CUB, Stanford Dogs, Stanford Cars, and Aircraft). As shown in Tables I, II, III, IV, and Tables V–VIII, URN achieves strong performance under both the 1-shot and 5-shot settings across different datasets and image domains.

Compared with the FRN baseline, URN provides consistent improvements on both remote sensing and natural fine-grained benchmarks. In particular, URN achieves notable gains over existing reconstruction-based methods on FGSCR and remains competitive on NWPU-RESISC45, AID, and WHU-RS19. Similar improvements are also observed on CUB, Stanford Dogs,

TABLE V
PERFORMANCE ON THE CUB DATASET USING BOUNDING-BOX CROPPED IMAGES AS INPUT.

Method	Backbone	1-shot	5-shot
MatchNet [2]	Conv-4	67.73 ± 0.22	79.00 ± 0.15
ProtoNet [1]	Conv-4	63.73 ± 0.23	81.50 ± 0.16
FEAT [68]	Conv-4	68.87 ± 0.21	82.90 ± 0.14
DSN [50]	Conv-4	66.01 ± 0.20	85.41 ± 0.13
PRSN [33]	Conv-4	66.28 ± 0.24	80.10 ± 0.15
CTX [7]	Conv-4	69.64 ± 0.22	87.31 ± 0.14
FRN [6]	Conv-4	73.48 ± 0.20	88.43 ± 0.12
TDM [17]	Conv-4	74.39 ± 0.19	88.89 ± 0.12
FicNet [20]	Conv-4	75.27 ± 0.18	88.48 ± 0.13
Ours	Conv-4	**75.90 ± 0.18**	**89.76 ± 0.12**
MatchNet [2]	ResNet-12	71.87 ± 0.22	85.08 ± 0.14
ProtoNet [1]	ResNet-12	66.09 ± 0.24	82.50 ± 0.15
FEAT [68]	ResNet-12	73.27 ± 0.20	85.77 ± 0.13
DeepEMD [12]	ResNet-12	75.65 ± 0.19	88.69 ± 0.12
RENet [16]	ResNet-12	79.49 ± 0.18	91.11 ± 0.12
DSN [50]	ResNet-12	80.51 ± 0.18	90.23 ± 0.12
CTX [7]	ResNet-12	80.67 ± 0.18	91.55 ± 0.12
DeepBDC [10]	ResNet-12	83.55 ± 0.17	93.82 ± 0.11
FRN [6]	ResNet-12	83.16 ± 0.17	92.42 ± 0.11
TDM [17]	ResNet-12	83.36 ± 0.17	92.80 ± 0.11
FGFL [11]	ResNet-12	80.77 ± 0.18	92.01 ± 0.11
BiFRN [24]	ResNet-12	82.88 ± 0.17	92.84 ± 0.11
LCCRN [25]	ResNet-12	82.97 ± 0.17	93.63 ± 0.11
FicNet [20]	ResNet-12	80.97 ± 0.18	93.17 ± 0.11
RaPSPNet [21]	ResNet-12	76.64 ± 0.19	92.73 ± 0.12
TST_MFL [36]	ResNet-12	80.79 ± 0.18	91.15 ± 0.12
SRN [27]	ResNet-12	83.82 ± 0.17	93.45 ± 0.11
SRE [34]	ResNet-12	84.34 ± 0.17	92.42 ± 0.11
CutRot [37]	ResNet-12	83.92 ± 0.17	92.77 ± 0.11
SRML [35]	ResNet-12	83.05 ± 0.17	92.74 ± 0.11
Ours	ResNet-12	**85.43 ± 0.16**	**94.16 ± 0.10**

TABLE VI
PERFORMANCE ON STANFORD DOGS DATASET.

Method	Backbone	1-shot	5-shot
FOT [69]	Conv-4	49.32 ± 0.74	68.18 ± 0.69
TOAN [70]	Conv-4	49.30 ± 0.77	67.17 ± 0.49
DLG [71]	Conv-4	47.77 ± 0.86	67.07 ± 0.72
FRN [6]	Conv-4	62.66 ± 0.20	80.58 ± 0.12
TDM [17]	Conv-4	57.64 ± 0.22	75.77 ± 0.16
Ours	Conv-4	**63.78 ± 0.20**	**81.79 ± 0.12**
ProtoNet [1]	ResNet-12	73.81 ± 0.21	87.39 ± 0.12
DeepEMD [12]	ResNet-12	70.38 ± 0.30	85.24 ± 0.18
CTX [7]	ResNet-12	73.22 ± 0.22	85.90 ± 0.13
HelixFormer [72]	ResNet-12	65.92 ± 0.49	80.65 ± 0.36
TST_MFL [36]	ResNet-12	67.84 ± 0.82	81.72 ± 0.61
FRN [6]	ResNet-12	76.49 ± 0.21	88.22 ± 0.12
BiFRN [24]	ResNet-12	72.54 ± 0.49	85.86 ± 0.28
LCCRN [25]	ResNet-12	75.87 ± 0.20	88.36 ± 0.11
C2-Net [23]	ResNet-12	75.50 ± 0.49	87.65 ± 0.28
FicNet [20]	ResNet-12	72.41 ± 0.64	85.11 ± 0.37
SRN [27]	ResNet-12	76.54 ± 0.21	88.52 ± 0.12
SRML [35]	ResNet-12	72.97 ± 0.47	86.01 ± 0.30
AIS-MLI [22]	ResNet-12	76.32 ± 0.47	88.25 ± 0.27
CutRot [37]	ResNet-12	77.88 ± 0.21	89.05 ± 0.12
Ours	ResNet-12	**78.98 ± 0.20**	**89.72 ± 0.12**

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TABLE VII
PERFORMANCE ON STANFORD CARS DATASET.

Method	Backbone	1-shot	5-shot
FOT [69]	Conv-4	54.55 ± 0.73	73.69 ± 0.65
TOAN [70]	Conv-4	65.90 ± 0.72	84.24 ± 0.48
DLG [71]	Conv-4	62.56 ± 0.82	88.98 ± 0.47
FRN [6]	Conv-4	69.72 ± 0.20	88.93 ± 0.12
TDM [17]	Conv-4	68.36 ± 0.22	86.14 ± 0.13
Ours	Conv-4	**70.18 ± 0.20**	**89.88 ± 0.12**
ProtoNet [1]	ResNet-12	85.46 ± 0.19	95.08 ± 0.12
DeepEMD [12]	ResNet-12	80.62 ± 0.26	92.63 ± 0.13
CTX [7]	ResNet-12	85.03 ± 0.19	92.63 ± 0.11
HelixFormer [72]	ResNet-12	79.40 ± 0.43	92.26 ± 0.15
TST_MFL [36]	ResNet-12	82.82 ± 0.84	93.61 ± 0.35
FRN [6]	ResNet-12	86.28 ± 0.17	94.78 ± 0.07
BiFRN [24]	ResNet-12	87.80 ± 0.17	96.34 ± 0.07
LCCRN [25]	ResNet-12	87.04 ± 0.17	96.19 ± 0.17
FicNet [20]	ResNet-12	86.81 ± 0.47	95.36 ± 0.22
RaPSPNet [21]	ResNet-12	78.84 ± 0.69	93.79 ± 0.49
C2-Net [23]	ResNet-12	88.00 ± 0.40	94.52 ± 0.23
SRN [27]	ResNet-12	88.02 ± 0.16	96.23 ± 0.07
SRML [35]	ResNet-12	87.49 ± 0.36	95.34 ± 0.16
Ours	ResNet-12	**89.10 ± 0.17**	**97.00 ± 0.08**

TABLE VIII
PERFORMANCE ON AIRCRAFT DATASET.

Method	Backbone	1-shot	5-shot
ProtoNet [1]	Conv-4	47.72 ± 0.24	69.42 ± 0.18
DSN [50]	Conv-4	49.63 ± 0.23	66.36 ± 0.17
CTX [7]	Conv-4	49.67 ± 0.23	69.06 ± 0.17
FRN [6]	Conv-4	53.20 ± 0.21	71.17 ± 0.15
TDM [17]	Conv-4	54.21 ± 0.21	71.37 ± 0.15
Ours	Conv-4	**54.58 ± 0.21**	**74.99 ± 0.14**
ProtoNet [1]	ResNet-12	67.28 ± 0.20	83.21 ± 0.13
DSN [50]	ResNet-12	70.23 ± 0.19	83.05 ± 0.13
CTX [7]	ResNet-12	65.53 ± 0.21	79.31 ± 0.15
FRN [6]	ResNet-12	69.58 ± 0.20	82.98 ± 0.13
TDM [17]	ResNet-12	70.89 ± 0.19	84.54 ± 0.12
C2-Net [23]	ResNet-12	71.39 ± 0.19	83.33 ± 0.12
SRN [27]	ResNet-12	67.34 ± 0.20	84.06 ± 0.12
Ours	ResNet-12	**72.92 ± 0.18**	**85.87 ± 0.11**

TABLE IX
PERFORMANCE ON MINI-IMAGENET.

Method	Backbone	1-shot	5-shot
MatchNet [2]	ResNet-12	65.64 ± 0.20	78.72 ± 0.15
ProtoNet [1]	ResNet-12	62.39 ± 0.21	80.53 ± 0.14
MetaOptNet [73]	ResNet-12	62.64 ± 0.61	78.63 ± 0.46
SimpleShot [74]	ResNet-12	62.85 ± 0.20	80.02 ± 0.14
CAN [15]	ResNet-12	63.85 ± 0.48	79.44 ± 0.34
DSN [12]	ResNet-12	62.64 ± 0.66	78.83 ± 0.45
FEAT [68]	ResNet-12	66.78 ± 0.20	82.05 ± 0.14
DeepEMD [12]	ResNet-12	65.91 ± 0.82	82.41 ± 0.56
FRN [6]	ResNet-12	66.45 ± 0.19	82.83 ± 0.13
LCCRN [25]	ResNet-12	62.24 ± 0.20	79.19 ± 0.14
Ours	ResNet-12	**66.98 ± 0.20**	**83.31 ± 0.19**

Stanford Cars, and Aircraft, demonstrating the robustness of the proposed framework across diverse few-shot classification scenarios.

These improvements mainly benefit from the proposed two-stage reconstruction mechanism, which alleviates representation and reconstruction biases in conventional one-stage reconstruction. By introducing inverse reconstruction and cyclic consistency between support and query representations, URN encourages more reliable feature reconstruction and similarity estimation, leading to consistent gains across different datasets.

2) Few-Shot General Classification: Unlike the fine-grained benchmarks used as the primary evaluation setting, mini-ImageNet is a general few-shot classification benchmark. Therefore, we use it as a supplementary experiment to evaluate whether URN can generalize beyond fine-grained recognition. As shown in Table IX, URN achieves 66.98% and 83.31% under the 1-shot and 5-shot settings, respectively. Compared with FRN, URN improves the accuracy by 0.53 and 0.48 percentage points, while also outperforming another reconstruction-based method, LCCRN. These results indicate that the proposed two-stage reconstruction framework remains effective in conventional few-shot classification settings.

The improvement on mini-ImageNet is relatively smaller than that observed on several fine-grained benchmarks. A possible explanation is that general few-shot classification usually exhibits larger inter-class differences, making classification less dependent on subtle local details. Consequently, the adverse effects of representation and reconstruction biases may be less pronounced. In contrast, fine-grained recognition relies more heavily on preserving subtle discriminative cues and accurately modeling support–query relationships, making the proposed cyclic reconstruction mechanism more beneficial.

Overall, these results suggest that URN is particularly effective for fine-grained recognition while still maintaining good generalization ability on conventional few-shot classification tasks.

C. Ablation Study

In this section, we conduct comprehensive ablation studies to evaluate the contribution of key components in our framework, including the inverse reconstruction constraint, the iteration number, the initialization strategy of $\boldsymbol{H}$, the regularization parameter, and the choice of optimization algorithm. We further examine the effect of replacing the CNN backbone with a Transformer-based one, and assess the effectiveness of extending the cyclic reconstruction mechanism to non-reconstruction methods.

1) Effect of the Inverse Reconstruction Constraint: Compared to the baseline FRN, URN introduces an inverse reconstruction constraint to explicitly address the issues of representation bias and reconstruction bias. Figure 3 compares FRN and URN on four datasets in 1-shot and 5-shot scenarios with the ResNet-12 backbone. As shown, URN achieves an average accuracy improvement of 2.73% in the 1-shot scenario and 2.08% in the 5-shot scenario over FRN.

The performance gains are not simply the result of an additional regularization term, but stem from the theoretical role of the inverse reconstruction constraint in shaping the learned projection matrix. In FRN, the mapping from support to query features is learned in a one-stage manner, making it prone to overfitting to domain- or class-specific structures present in the base classes. This leads to *representation bias*, where the learned projection captures patterns that fail

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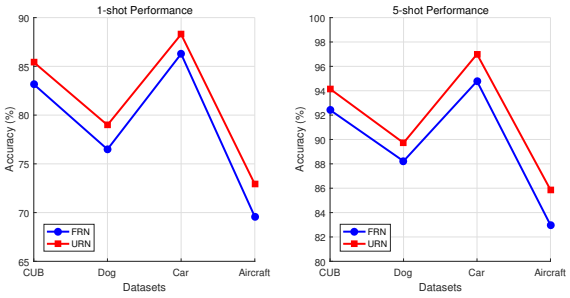


Fig. 3. The effect of the inverse reconstruction constraint.

TABLE X
COMPUTATIONAL COST OF URN WITH DIFFERENT NUMBERS OF ALTERNATING ITERATIONS UNDER THE 5-WAY 1-SHOT SETTING USING THE RESNET-12 BACKBONE. THE FIRST ROW DENOTES THE FRN BASELINE. SINCE ALL DATASETS SHARE THE SAME INPUT RESOLUTION AND EPISODIC CONFIGURATION, THE REPORTED COMPUTATIONAL COST IS DATASET-INDEPENDENT.

Iterations	Latency	FLOPs	Params
–	34.48 ms	105.93 G	12.4 M
1	34.50 ms	105.99 G	12.4 M
2	34.69 ms	106.03 G	12.4 M
3	34.77 ms	106.06 G	12.4 M
4	34.96 ms	106.09 G	12.4 M
5	35.34 ms	106.12 G	12.4 M
6	35.57 ms	106.15 G	12.4 M
7	35.79 ms	106.18 G	12.4 M
8	36.05 ms	106.22 G	12.4 M

to generalize to novel categories. Furthermore, because the projection is optimized only in the forward direction, systematic *reconstruction bias* can occur, especially when the feature distributions of support and query sets differ.

The inverse reconstruction constraint mitigates these issues by enforcing bidirectional consistency: a feature projected from the support to the query space must, when re-projected back, recover the original support representation with minimal error. This two-way consistency acts as an implicit regularizer, constraining the projection to preserve intrinsic geometric relationships in the feature space rather than overfitting to specific domain artifacts. As a result, the learned projection matrix becomes both more stable and more domain-invariant, enabling URN to produce more discriminative and robust feature reconstructions. The consistent improvements across datasets and shot settings confirm that this constraint effectively reduces bias and enhances generalization in few-shot classification.

2) Effect of the Number of Iterations: To investigate the effect of the alternating optimization depth, we vary the number of iterations N from 1 to 8. We separately analyze the computational overhead and classification performance to avoid duplicated information. Table X reports the per-episode latency, FLOPs, and parameter count under different numbers of alternating iterations, while Table XI reports the corresponding 5-way 1-shot accuracy on CUB and four remote sensing datasets, including FGSCR, NWPU-RESISC45, AID, and WHU-RS19.

Under our unified episodic evaluation protocol, all datasets

TABLE XI
EFFECT OF THE NUMBER OF ALTERNATING ITERATIONS N ON 1-SHOT CLASSIFICATION ACCURACY ACROSS CUB AND FOUR REMOTE SENSING DATASETS USING THE RESNET-12 BACKBONE.

Iterations	CUB	FGSCR	RESISC45	AID	WHU-RS19
1	85.14	59.82	69.81	65.94	84.31
2	85.20	60.03	69.97	66.08	84.46
3	84.88	60.11	70.06	66.15	84.58
4	85.08	60.27	70.17	66.27	84.73
5	**85.43**	**60.36**	**70.23**	**66.36**	**84.88**
6	85.11	60.31	70.19	66.31	84.82
7	85.18	60.28	70.16	66.28	84.79
8	85.15	60.24	70.14	66.25	84.76

TABLE XII
EFFECT OF INITIAL VALUES OF $\boldsymbol{H}$ ON 1-SHOT AND 5-SHOT ACCURACY FOR CONV-4 AND RESNET-12 BACKBONES ON THE CUB DATASET.

Initial Value $\boldsymbol{H}$	Conv-4		ResNet-12	
	1-shot	5-shot	1-shot	5-shot
$\boldsymbol{N}$	73.71	89.19	85.28	93.98
$\boldsymbol{I}$	73.51	88.93	85.33	94.00
$\boldsymbol{O}$	**75.90**	**89.76**	**85.43**	**94.16**

use the same 84×84 input resolution, ResNet-12 backbone, number of ways, number of query samples, and feature-map dimensions. Therefore, for a fixed N, the dimensions of $\boldsymbol{S}_c$, $\boldsymbol{Q}$, $\boldsymbol{W}$, and $\boldsymbol{H}$ remain unchanged across datasets. As a result, the theoretical FLOPs and parameter count are identical for all datasets. We further measured the actual per-episode inference latency on all evaluated datasets under the same hardware configuration and observed only negligible runtime fluctuations, confirming that the computational cost is essentially dataset-independent.

As shown in Table X, increasing the number of alternating iterations introduces only marginal additional computational cost. The parameter count remains unchanged at 12.4 M because the iterative updates do not introduce additional learnable modules. The FLOPs increase only from 105.99 G at $N = 1$ to 106.12 G at $N = 5$, while the measured per-episode latency increases from 34.50 ms to 35.34 ms. Even when N is further increased to 8, the latency and FLOPs remain only 36.05 ms and 106.22 G, respectively.

Table XI further shows that the accuracy trend is highly consistent across datasets. The performance generally improves during the first several alternating updates and reaches the best or a stable value around $N = 5$ on CUB, FGSCR, NWPU-RESISC45, AID, and WHU-RS19. Further increasing the number of iterations provides little additional benefit and may even lead to slight performance fluctuations.

These results indicate that the choice of $N = 5$ is not specific to a single dataset. More importantly, the improvement in classification accuracy is achieved with only a very small increase in latency and FLOPs, demonstrating a favorable computation–accuracy tradeoff of URN for both natural-image and remote sensing few-shot classification. Accordingly, we set $N = 5$ for all subsequent experiments.

3) Effect of the Initial Value of $\boldsymbol{H}$: The alternating optimization algorithm in URN requires an initial value for $\boldsymbol{H}$.

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TABLE XIII
PERFORMANCE COMPARISON OF DIFFERENT FIXED AND LEARNABLE SETTINGS OF THE FOUR REGULARIZATION PARAMETERS λ_o, λ_d, λ_c, AND λ_e ON THE CUB DATASET. FOR THE FIXED-PARAMETER EXPERIMENTS, THE REMAINING PARAMETERS ARE LEARNABLE.

λ_o	λ_d	λ_c	λ_e	Conv-4		ResNet-12	
				1-shot	5-shot	1-shot	5-shot
Learned	Learned	0.1	0.1	71.94	89.83	84.77	93.98
Learned	Learned	0.2	0.2	71.48	88.96	84.62	93.96
Learned	Learned	0.3	0.3	72.50	89.40	84.68	94.09
Learned	Learned	0.4	0.4	71.75	89.09	84.65	93.98
Learned	Learned	0.1	0.2	72.18	89.54	84.81	94.00
Learned	Learned	0.1	0.3	71.86	89.26	84.73	94.04
Learned	Learned	0.2	0.1	72.46	89.67	84.86	94.02
Learned	Learned	0.2	0.3	72.21	89.18	84.74	94.07
Learned	Learned	0.3	0.1	72.83	89.71	84.91	94.08
Learned	Learned	0.3	0.2	73.06	89.68	84.96	94.11
0.05	0.05	Learned	Learned	74.82	89.48	85.02	94.02
0.10	0.10	Learned	Learned	75.21	89.62	85.18	94.08
0.20	0.20	Learned	Learned	74.93	89.55	85.08	94.06
0.30	0.30	Learned	Learned	74.51	89.37	84.92	93.99
0.10	0.20	Learned	Learned	75.34	89.66	85.22	94.10
0.20	0.10	Learned	Learned	75.08	89.61	85.14	94.07
Learned	Learned	Learned	Learned	**75.90**	**89.76**	**85.43**	**94.16**

We evaluated three initialization strategies: a zero matrix $\boldsymbol{O}$, an identity matrix $\boldsymbol{I}$, and a standard normal distribution matrix $\boldsymbol{N}$. As reported in Table XII, performance differences are modest but consistent, with the zero initialization $\boldsymbol{O}$ yielding the highest accuracy in both Conv-4 and ResNet-12 backbones.

This result can be explained from the perspective of cyclic reconstruction optimization. When $\boldsymbol{H}$ is initialized to zeros, the first iteration starts from a neutral state with no prior directional bias in the projection space. This forces the updates in both forward and inverse projections to be driven entirely by the observed support–query relationships in the current task, leading to a more balanced and domain-agnostic refinement process. In contrast, identity initialization $\boldsymbol{I}$ implicitly assumes that the initial forward projection should preserve the original feature coordinates, which may embed base-domain-specific geometry into the first iteration and thus slow adaptation to novel categories. Similarly, random initialization $\boldsymbol{N}$ introduces stochastic biases in the initial projection, potentially causing unstable early updates that need additional iterations to converge.

From an optimization standpoint, zero initialization also minimizes the risk of early overfitting: because $\boldsymbol{H}$ enters the cyclic constraint in both forward and inverse directions, any biased starting point can propagate through the iterations, accumulating representation and reconstruction biases before the refinement process stabilizes. By starting from $\boldsymbol{O}$, URN ensures that both projections evolve symmetrically from an unbiased baseline, allowing the bidirectional constraints to more effectively guide convergence toward a well-regularized projection matrix. Based on this combination of empirical performance and theoretical stability, we adopt $\boldsymbol{O}$ as the default initialization for $\boldsymbol{H}$ in all experiments.

4) Effect of the Learned Regularization Parameters: URN contains four regularization coefficients, λ_o, λ_c, λ_d, and λ_e, which control different terms in the forward and inverse reconstruction objectives. To systematically evaluate their influence and justify the learnable regularization strategy, we conduct two complementary experiments: fixed-value sensitivity analysis and initialization sensitivity analysis.

TABLE XIV
SENSITIVITY ANALYSIS OF DIFFERENT INITIAL VALUES FOR THE FOUR LEARNABLE REGULARIZATION PARAMETERS λ_o, λ_c, λ_d, AND λ_e ON THE CUB DATASET.

$\lambda_o^{(0)}$	$\lambda_c^{(0)}$	$\lambda_d^{(0)}$	$\lambda_e^{(0)}$	Conv-4		ResNet-12	
				1-shot	5-shot	1-shot	5-shot
0.01	0.01	0.01	0.01	75.61	89.65	85.25	94.07
0.05	0.05	0.05	0.05	75.74	89.70	85.34	94.11
0.10	0.10	0.10	0.10	**75.90**	**89.76**	**85.43**	**94.16**
0.20	0.20	0.20	0.20	75.78	89.72	85.36	94.13
0.30	0.30	0.30	0.30	75.65	89.67	85.29	94.09

First, we investigate the effect of manually fixing different regularization coefficients. As summarized in Table XIII, we consider two groups of experiments. In the first group, λ_o and λ_d are kept learnable, while λ_c and λ_e are assigned different fixed combinations. In the second group, λ_c and λ_e are kept learnable, while different fixed combinations of λ_o and λ_d are evaluated.

For λ_c and λ_e, the fixed configuration $\lambda_c = 0.3$ and $\lambda_e = 0.2$ achieves the strongest overall performance among the tested combinations, reaching 73.06%/89.68% with Conv-4 and 84.96%/94.11% with ResNet-12 under the 1-shot/5-shot settings. For λ_o and λ_d, the best fixed combination is $\lambda_o = 0.1$ and $\lambda_d = 0.2$, which achieves 75.34%/89.66% with Conv-4 and 85.22%/94.10% with ResNet-12.

Although carefully selected fixed coefficients can provide competitive performance, their optimal values vary across different regularization terms, backbones, and shot settings. In contrast, when all four parameters are jointly learned, URN achieves the best overall results: 75.90%/89.76% with Conv-4 and 85.43%/94.16% with ResNet-12. These results indicate that jointly optimizing the regularization strengths can better adapt the balance among the different reconstruction and consistency objectives than using a single manually selected configuration.

We further analyze whether the learnable strategy is sensitive to the initial values of the regularization parameters. Specifically, all four parameters are initialized with the same value selected from $\{0.01, 0.05, 0.10, 0.20, 0.30\}$ and are then jointly optimized during training. The results are reported in Table XIV. For Conv-4, the 1-shot accuracy varies only from 75.61% to 75.90%, while the 5-shot accuracy ranges from 89.65% to 89.76%. Similarly, for ResNet-12, the 1-shot accuracy ranges from 85.25% to 85.43%, and the 5-shot accuracy ranges from 94.07% to 94.16%.

The relatively small variation across different initializations shows that the learnable regularization strategy does not rely on a specific starting value. More importantly, it avoids exhaustive manual tuning while consistently achieving stronger overall performance than the evaluated fixed configurations. Therefore, we adopt jointly learnable λ_o, λ_c, λ_d, and λ_e in all subsequent experiments.

5) Comparison of Different Optimization Methods: The cyclic reconstruction objective in URN leads to a coupled matrix optimization problem, whose first-order optimality

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TABLE XV
COMPARISON OF DIFFERENT OPTIMIZATION STRATEGIES ON CUB WITH THE RESNET-12 BACKBONE. "N/A" INDICATES THAT NO CONVERGED CLASSIFICATION ACCURACY CAN BE OBTAINED BECAUSE TRAINING FAILS DUE TO NUMERICAL INSTABILITY.

Optimization	1-shot	5-shot	Latency	Stability	Training
Bartels–Stewart	N/A	N/A	1420 ms	Unstable	Failed
Our proposed	**85.43**	**94.16**	**35 ms**	Stable	Successful

condition with respect to the projection matrix can be expressed as a Sylvester equation. One possible solution is to directly apply a classical Sylvester-equation solver, such as the Bartels–Stewart algorithm. Alternatively, our method introduces an auxiliary variable $\boldsymbol{H}$ and adopts an alternating optimization strategy, which decomposes the original coupled problem into two regularized least-squares subproblems with closed-form updates.

To evaluate the necessity of this reformulation, we directly implement the Bartels–Stewart algorithm as an additional baseline without using the auxiliary variable $\boldsymbol{H}$. The comparison is reported in Table XV. Although the direct solver can produce valid projection matrices when applied to fixed feature representations at inference time, it exhibits severe numerical instability during end-to-end episodic training. In particular, we repeatedly observe numerical overflow, NaN values, and exploding gradients before convergence. Consequently, the direct solver fails to complete training and no meaningful converged classification accuracy can be obtained.

In contrast, the proposed alternating optimization successfully completes end-to-end training and achieves 85.43% and 94.16% accuracy under the 5-way 1-shot and 5-way 5-shot settings, respectively. It also requires only approximately 35 ms per episode, whereas the direct Bartels–Stewart solver requires about 1420 ms, corresponding to more than a $40\times$ increase in latency.

The difference mainly arises from the optimization form. The Bartels–Stewart algorithm relies on Schur decomposition followed by triangular equation solving, and backpropagation through these matrix operations becomes numerically fragile when the coefficient matrices are poorly conditioned. Such situations can occur frequently in few-shot episodes because only a limited number of support samples are available and their feature representations can be highly correlated. By introducing $\boldsymbol{H}$, the proposed formulation avoids directly differentiating through the large-scale Sylvester solver and instead alternates between two regularized closed-form subproblems.

Therefore, the introduction of the auxiliary variable $\boldsymbol{H}$ is not merely intended to reduce computational cost. More importantly, it makes the cyclic reconstruction objective substantially more stable and practical for end-to-end few-shot learning, while preserving the closed-form nature of each optimization step.

6) Transformer Backbone: To further evaluate the backbone generality of URN, we replace the CNN backbone with ViT-S and conduct experiments on FGSCR, NWPU-RESISC45, AID, and WHU-RS19. As shown in Table XVI, URN consistently outperforms FRN under all evaluated 1-shot and 5-shot settings. The average improvements over FRN are 1.88 and 3.17 percentage points for the 1-shot and 5-shot settings, respectively. These results demonstrate that the proposed two-stage reconstruction framework is not restricted to CNN-based features and can also be effectively applied to Transformer-based representations.

7) Generalization of the Cyclic Reconstruction Mechanism: To further evaluate whether the proposed cyclic reconstruction mechanism generalizes beyond conventional within-dataset base–novel splits, we conduct additional cross-domain few-shot classification experiments using different remote sensing datasets as source and target domains.

Specifically, we consider two independent source-domain settings. In the first setting, AID is used as the source dataset, and the trained model is directly evaluated on WHU-RS19, NWPU-RESISC45, and FGSCR. In the second setting, FGSCR is used as the source dataset, and the model is directly evaluated on WHU-RS19, AID, and NWPU-RESISC45. In all experiments, the model is trained only on the source dataset, while no target-domain fine-tuning or adaptation is performed. Evaluation is conducted under both 5-way 1-shot and 5-way 5-shot settings following the same episodic protocol as in the within-domain experiments.

This evaluation covers different levels of cross-domain discrepancy. Transfers among AID, NWPU-RESISC45, and WHU-RS19 mainly reflect dataset-level differences among remote sensing scene datasets. In contrast, transfers involving FGSCR represent a more heterogeneous cross-dataset setting because FGSCR focuses on fine-grained ship recognition and therefore differs more substantially from the scene-classification datasets in both image content and semantic granularity. The corresponding results are reported in Table XVII. When AID is used as the source domain, URN achieves 92.17% and 97.88% on WHU-RS19, 63.81% and 82.55% on NWPU-RESISC45, and 42.39% and 71.13% on FGSCR under the 1-shot and 5-shot settings, respectively. When FGSCR is used as the source domain, URN achieves 68.85% and 88.01% on WHU-RS19, 53.71% and 74.40% on AID, and 52.85% and 73.53% on NWPU-RESISC45, respectively. Compared with the one-stage reconstruction baseline FRN, URN consistently achieves higher accuracy across all evaluated source–target combinations and under both shot settings. This result indicates that the benefit of cyclic reconstruction is not restricted to the conventional within-dataset setting. Even when the feature distribution shifts across datasets, the inverse reconstruction constraint continues to provide a more reliable support–query mapping. A possible explanation is that cross-domain shifts increase the discrepancy between support and query representations, making one-way reconstruction more susceptible to information loss and reconstruction mismatch. By additionally requiring the projected representation to remain recoverable through the inverse mapping, URN imposes a bidirectional structural constraint that discourages excessive suppression of transferable information and improves the robustness of support–query reconstruction. Nevertheless, the absolute accuracy varies considerably across different source–target pairs. In particular, the heterogeneous transfers involving FGSCR are substantially more challenging than transfers among remote sensing scene datasets. Therefore, although cyclic reconstruction exhibits

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TABLE XVI
PERFORMANCE COMPARISON BETWEEN FRN AND URN USING THE VIT-S BACKBONE ON FOUR REMOTE SENSING DATASETS.

Setting	Method	Backbone	FGSCR	NWPU-RESISC45	AID	WHU-RS19
1-shot	FRN	ViT-S	55.64 ± 0.17	68.57 ± 0.21	64.36 ± 0.21	81.21 ± 0.14
	URN	ViT-S	**57.17 ± 0.17**	**69.58 ± 0.20**	**66.60 ± 0.20**	**83.95 ± 0.14**
5-shot	FRN	ViT-S	79.19 ± 0.11	83.83 ± 0.12	80.28 ± 0.15	93.19 ± 0.06
	URN	ViT-S	**82.21 ± 0.11**	**87.48 ± 0.11**	**84.70 ± 0.11**	**94.76 ± 0.05**

TABLE XVII
CROSS-DOMAIN FEW-SHOT REMOTE SENSING CLASSIFICATION RESULTS. MODELS ARE TRAINED ONLY ON THE SOURCE DATASET AND DIRECTLY EVALUATED ON THE TARGET DATASET WITHOUT TARGET-DOMAIN FINE-TUNING OR ADAPTATION.

Source	Target	Method	1-shot		5-shot	
			Accuracy	Δ	Accuracy	Δ
AID	WHU-RS19	FRN	90.84 ± 0.11	–	97.12 ± 0.05	–
		URN	**92.17 ± 0.09**	**+1.34**	**97.88 ± 0.03**	**+0.76**
	NWPU-RESISC45	FRN	61.64 ± 0.22	–	80.72 ± 0.14	–
		URN	**63.81 ± 0.20**	**+2.18**	**82.55 ± 0.13**	**+1.84**
	FGSCR	FRN	40.27 ± 0.16	–	68.94 ± 0.14	–
		URN	**42.39 ± 0.14**	**+2.13**	**71.13 ± 0.12**	**+2.20**
FGSCR	WHU-RS19	FRN	66.42 ± 0.17	–	86.25 ± 0.09	–
		URN	**68.85 ± 0.15**	**+2.43**	**88.01 ± 0.08**	**+1.77**
	AID	FRN	51.46 ± 0.21	–	72.31 ± 0.16	–
		URN	**53.71 ± 0.19**	**+2.26**	**74.40 ± 0.14**	**+2.10**
	NWPU-RESISC45	FRN	50.44 ± 0.22	–	71.08 ± 0.17	–
		URN	**52.85 ± 0.20**	**+2.42**	**73.53 ± 0.15**	**+2.46**

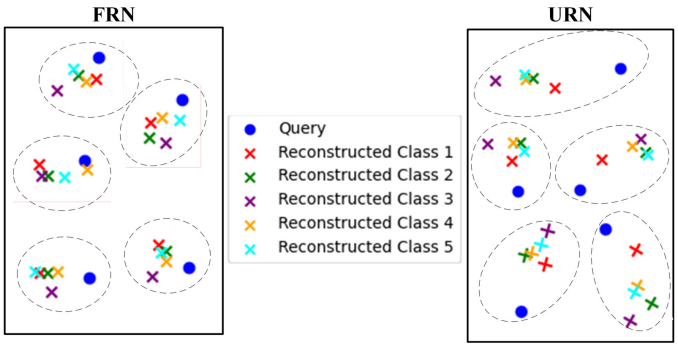


Fig. 4. Visualization results of reconstructed features. In our experiments, the query and reconstructed features from class 1 belong to the same class

favorable cross-domain generalization, it cannot completely eliminate severe distribution discrepancies. These results demonstrate the generalizability of the proposed reconstruction mechanism while also clarifying its practical boundary under large cross-domain shifts.

D. Qualitative Results

Feature Distribution and Reconstruction Analysis: To provide an intuitive comparison between URN and FRN, we first visualize the query features and reconstructed query features using t-SNE. Specifically, we randomly sample a 5-way 5-shot episode from the test set and select one query image for reconstruction. The resulting query feature map has a spatial resolution of 5×5 with 640 channels, corresponding to 25 local feature descriptors. For clarity, Figure 4 visualizes five representative query descriptors together with their reconstructed counterparts obtained from the five support classes. The query and the reconstructed features from class 1 belong to the same class, while the remaining reconstructed features correspond to different classes.

As shown in Figure 4, FRN does not always place the same-class reconstructed feature closest to the corresponding query feature. In several cases, reconstructed features from different classes are located closer to the query, indicating that one-stage reconstruction may distort discriminative relationships because of representation and reconstruction biases. In contrast, URN produces a more discriminative relative arrangement of reconstructed features: same-class reconstructions are more consistently associated with the query, while different-class reconstructions are better separated. This observation qualitatively supports the effectiveness of the inverse reconstruction constraint in preserving useful class-discriminative information during projection.

It should be noted that the absolute intra-class distance does not necessarily decrease after applying URN. The more important observation is the relative separation between intra-class and inter-class reconstructed features. In FRN, both distances can be relatively small, which causes different classes to overlap in the reconstructed feature space. After introducing cyclic reconstruction, the inter-class separation becomes more pronounced, while the intra-class variation is still preserved. Consequently, the relative margin between correct and incorrect reconstructed classes becomes larger, reducing ambiguity in the reconstruction-based metric space.

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To further verify this observation, we visualize all 25 local feature descriptors of the query feature map and their reconstructed counterparts. Figure 5 presents the result of FRN, whereas Figure 6 shows that of URN. Consistent with the sparse descriptor visualization, FRN exhibits considerable overlap between same-class and different-class reconstructed features. In comparison, URN produces a clearer separation between the correct class and competing classes, indicating improved support–query alignment and better class discriminability after cyclic reconstruction.

Failure Case Analysis: While the above visualizations demonstrate the effectiveness of URN in improving feature reconstruction and class separability, we further analyze representative failure cases to clarify the practical boundary of the proposed method. Figure 7 shows misclassified samples from the FGSCR dataset under the 5-way 1-shot setting. For each query, the complete support set is displayed, where the blue box denotes the support sample from the ground-truth class and the red box denotes the support sample from the predicted/confusing class.

The failure cases can be summarized into four representative patterns. First, degraded appearance caused by cloud cover or poor imaging quality may obscure important structural characteristics of the target. Second, low-contrast targets provide insufficient fine-grained visual cues, making different ship categories difficult to distinguish. Third, complex dock environments introduce strong background structures that may interfere with the limited target information. Finally, when local discriminative details are weak or blurred, the query representation may become more similar to a confusing class than to its ground-truth support sample.

These examples reveal an important limitation of URN. The proposed cyclic reconstruction mainly improves the preservation and recoverability of information during support–query projection, but it cannot recover discriminative cues that have already been severely degraded or lost during image acquisition or backbone feature extraction. Therefore, although URN improves the reliability of reconstruction-based classification, its performance remains dependent on the quality and discriminability of the underlying feature representation in extremely challenging fine-grained scenarios.

V. Conclusion

In this paper, we proposed the Unbiased Reconstruction Network (URN), a novel two-stage reconstruction framework for few-shot fine-grained remote sensing classification. URN addresses two key limitations of existing reconstruction-based methods, namely representation bias and reconstruction bias, by introducing a cyclic reconstruction objective with an inverse reconstruction constraint. To efficiently optimize the proposed objective, we introduced an auxiliary matrix variable and developed an alternating optimization algorithm with closed-form solutions. This yields an efficient and scalable optimization procedure. Extensive experiments on remote sensing and natural image benchmarks demonstrate that URN achieves strong overall performance, validating its effectiveness and generalizability for few-shot fine-grained classification tasks.

Despite these advantages, URN still has some limitations. The current formulation relies on linear projection and quadratic reconstruction objectives, which may limit its ability to model more complex nonlinear relationships between support and query features. In future work, we will explore more flexible nonlinear reconstruction mechanisms and extend URN to broader few-shot learning scenarios, such as object detection, and semantic segmentation.

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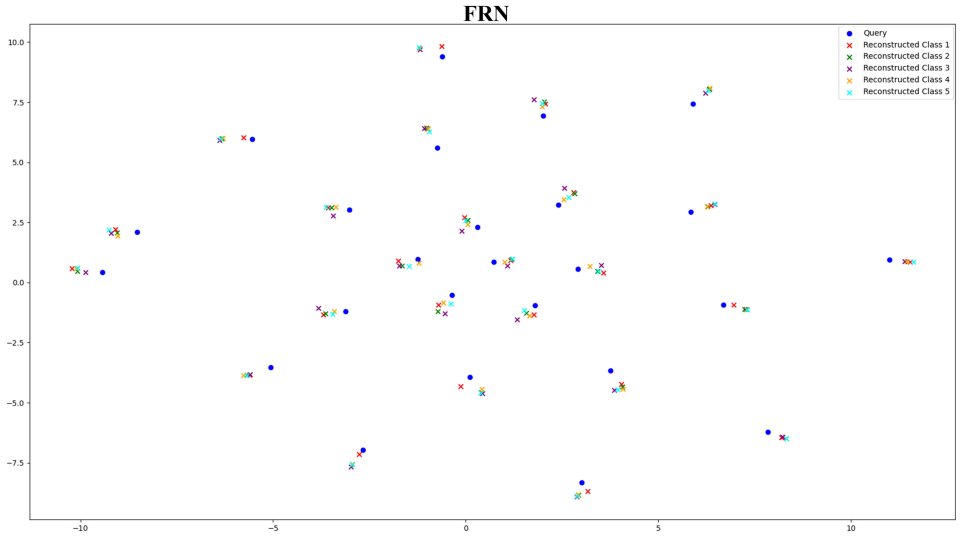


Fig. 5. Visualization results of reconstructed features in FRN. The query and reconstructed features from class 1 belong to the same class.

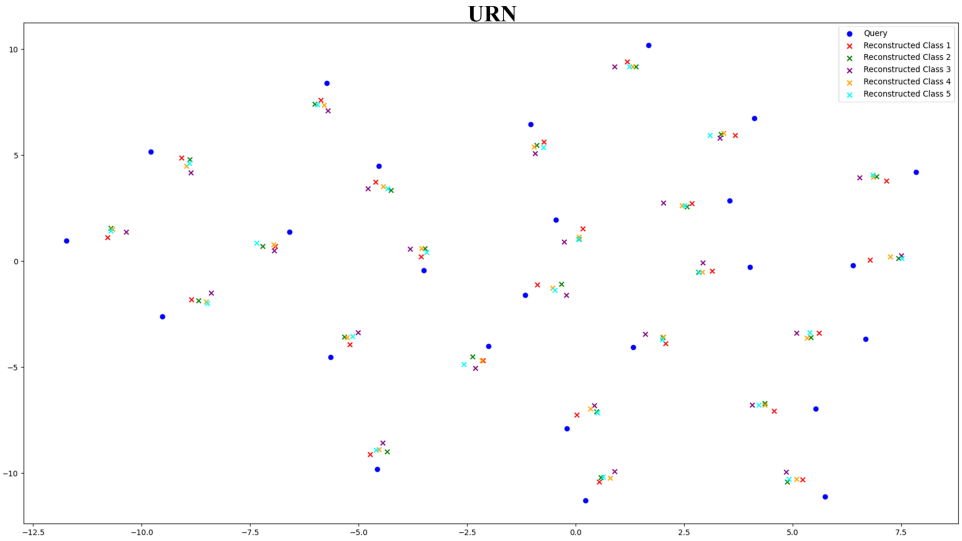


Fig. 6. Visualization results of reconstructed features in URN. The query and reconstructed features from class 1 belong to the same class.

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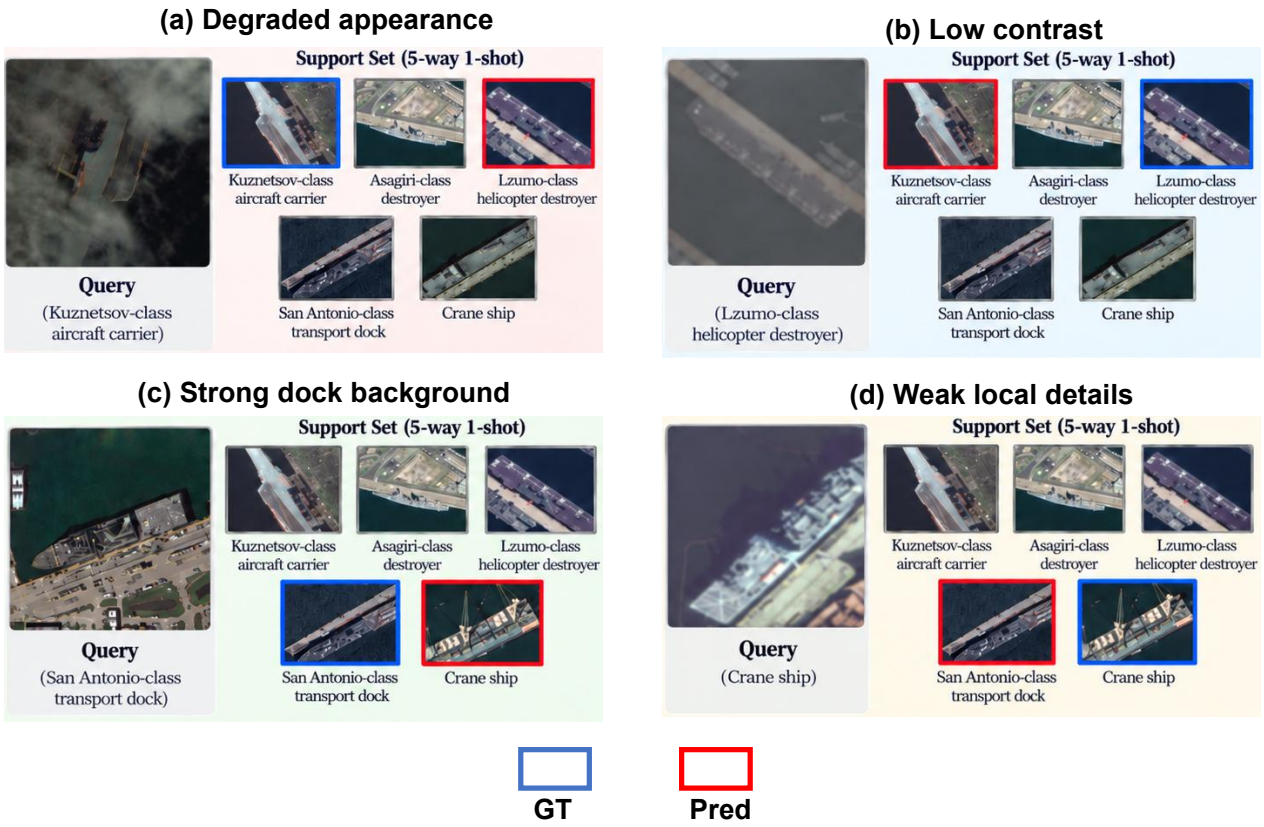


Fig. 7. Representative failure cases of URN on the FGSCR dataset under the 5-way 1-shot setting. Each case shows a query image together with its five support samples. The blue box denotes the support sample from the ground-truth class, while the red box denotes the support sample from the predicted/confusing class. Four representative failure patterns are illustrated: (a) degraded appearance, (b) low contrast, (c) strong dock background, and (d) weak local details.

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